

Nonuniqueness for Bounded Measurable Scalar Conductivities in Three Dimensions

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Abstract

We construct two distinct uniformly positive bounded measurable scalar conductivities on a ball in $\mathbb{R}^3$ with the same full Dirichlet-to-Neumann operator. Both conductivities equal one near the boundary. This gives nonuniqueness in the scalar Calderón problem at bounded measurable regularity.

1 Introduction

Let $\Omega \subset \mathbb{R}^n$ be a bounded connected Lipschitz domain and let $\gamma \in L^\infty(\Omega; \mathbb{R})$ satisfy $0 < c \leq \gamma \leq C < \infty$ almost everywhere. For $f \in H^{1/2}(\partial\Omega)$, let $u_f \in H^1(\Omega)$ be the unique function of trace f satisfying

$$\int_{\Omega} \gamma \nabla u_f \cdot \nabla \varphi \, dx = 0 \quad (\varphi \in H_0^1(\Omega)). \quad (1.1)$$

Its full weak Dirichlet-to-Neumann operator is defined by

$$\langle \Lambda_\gamma f, g \rangle = \int_{\Omega} \gamma \nabla u_f \cdot \nabla v \, dx, \quad \text{Tr } v = g. \quad (1.2)$$

Equation (1.1) makes this independent of the extension v . Our result concerns this entire operator on the usual trace space, at zero frequency. The inverse problem asks whether the boundary response to every applied voltage determines the conductivity inside the domain.

Theorem 1.1. *There exist real scalar functions $\gamma_0, \gamma_1 \in L^\infty(B(0, 3); \mathbb{R})$ and constants $0 < c < C < \infty$ such that*

- (i) $c \leq \gamma_j \leq C$ almost everywhere, for $j = 0, 1$;
- (ii) both conductivities equal 1 in a neighborhood of the boundary;
- (iii) $|\{x : \gamma_0(x) \neq \gamma_1(x)\}| > 0$;
- (iv) $\Lambda_{\gamma_0} = \Lambda_{\gamma_1}$ as bounded operators from $H^{1/2}(\partial B(0, 3))$ to $H^{-1/2}(\partial B(0, 3))$.

Thus the scalar Calderón uniqueness assertion with only bounded measurability and uniform positivity has a negative resolution, already in dimension three. In particular, the assertion quantified over every $n \geq 3$ is false. The theorem imposes no derivative regularity and does not prescribe the contrast C/c or assert a construction in every higher dimension.

Corollary 1.2 (Localized nonuniqueness). *Let c, C be the constants in Theorem 1.1. For every bounded connected Lipschitz domain $\Omega \subset \mathbb{R}^3$ and every finite family of pairwise disjoint balls $B_1, \dots, B_m \Subset \Omega$, $m \geq 1$, there are 2^m pairwise distinct real scalar conductivities $\gamma_\varepsilon \in L^\infty(\Omega)$, indexed by $\varepsilon \in \{0, 1\}^m$, such that*

$$c \leq \gamma_\varepsilon \leq C \quad \text{almost everywhere,} \quad \gamma_\varepsilon = 1 \quad \text{on } \Omega \setminus \bigcup_{i=1}^m B_i,$$

and all their full weak Dirichlet-to-Neumann operators from $H^{1/2}(\partial\Omega)$ to $H^{-1/2}(\partial\Omega)$ agree. Distinct conductivities differ on a set of positive measure.

The proof, given after the main construction, copies the pair into smaller separated balls and minimizes energy over their interface traces. It does not require the common response to be that of the constant conductivity one.

Regularity and related constructions. Calderón posed the determination problem for bounded measurable real scalar conductivities bounded away from zero, and proved injectivity of its linearization at constant conductivities [3]. For the nonlinear problem in dimensions at least three, Sylvester and Uhlmann established full-boundary uniqueness for smooth scalar conductivities [13]. Later results reach lower regularity. Haberman proves uniqueness for uniformly elliptic $W^{1,n}$ conductivities in dimensions three and four [6, Theorem 1.1]; Caro and Rogers prove it for positive Lipschitz conductivities in every dimension at least three [4, Theorem 1.1]. In the plane, Astala and Päiväranta prove uniqueness for bounded measurable uniformly positive scalar conductivities on bounded simply connected domains [2, Theorem 1]. Theorem 1.1 concerns the three-dimensional measurable class, without the derivative hypotheses of the cited higher-dimensional results.

The companion scalar corollary proves uniqueness for strictly positive conductivities smooth up to the boundary of a bounded connected smooth domain in $\mathbb{R}^n$, $n \geq 3$, with Dirichlet inputs supported in any nonempty relatively open boundary patch and conductivity flux observed on that same patch [10, Corollary 1.3].

For rough matrix-valued conductivities, nonunique continuation provides a route to inverse nonuniqueness. Daudé, Kamran, and Nicoleau [5, Theorem 1 and Proposition 1.2] combine Miller's construction [9] with a conformal change to obtain anisotropic counterexamples with partial boundary data. Their full-boundary obstruction [5, Remark 1.3] also explains a difficulty for the present problem: a weakly harmonic factor with constant full boundary trace is itself constant by the energy identity. The factor used below has a nonzero interior source, subject to a more specific annihilation identity.

Scalar approximation of matrix conductivities raises a separate issue. Marino and Spagnolo [8] establish isotropic density in the sense of G -convergence. Rondi, using an example communicated by Giovanni Alessandrini, makes the distinction between such approximation and exact boundary-map realization explicit [12, Proposition 2.2, Theorem 4.3, and Example 4.4]. On the unit disc, a constant matrix $\text{diag}(a, b)$ with $a, b > 0$ and $a \neq b$ is an H -limit of scalar coefficients, with ellipticity bounds allowed to widen, but has no bounded uniformly positive scalar representative with the same full boundary map. The scalarization used here preserves exact responses to two prescribed inputs; a further argument is needed to reach every input.

From a scalar block to the full operator. The construction in Section 5 takes $\Omega = B(0, 3)$ and nests scaled copies of a scalar block with one parent and two child boundary tori. Each torus carries a specified probability measure. The block can realize every current triple $p = (p_0, p_-, p_+)$ in

the plane $p_0 + p_- + p_+ = 0$, with flux equal to p_i times the corresponding measure. After its parent voltage is normalized to zero, the potential extends by constants into the adjacent regions as an admissible compactly supported test for any global γ -harmonic function u . Reciprocity therefore gives

$$p_0 m_0(u) + p_- m_-(u) + p_+ m_+(u) = 0 \quad (p_0 + p_- + p_+ = 0),$$

where $m_i(u)$ is the trace average on the indicated surface. The three averages are equal because their vector is orthogonal to the zero-sum plane. Iterating through the nested blocks gives one common average u_* on every cell surface, for every global harmonic u .

Signed currents propagated through the cells produce a nonzero $w \in H_0^1(\Omega) \cap L^\infty(\Omega)$ supported away from the outer boundary. The common averages and shrinking cell diameters imply that its source satisfies

$$\int_{\Omega} \gamma \nabla w \cdot \nabla ((u - u_*)\phi) \, dx = 0 \quad (\phi \in C_c^\infty(\Omega))$$

for every bounded weakly γ -harmonic u . The source is not zero; it vanishes on these particular products. This is the property used to change the conductivity.

Choose a sufficiently small nonzero δ so that $\rho = 1 + \delta w$ is positive, and set

$$\tilde{\gamma} = \rho^2 \gamma, \quad \tilde{u} = u_* + \frac{u - u_*}{\rho}.$$

The corresponding flux is

$$\tilde{\gamma} \nabla \tilde{u} = \rho \gamma \nabla u - (u - u_*) \gamma \nabla \rho.$$

The product identity, together with the original weak equation tested against $\rho\phi$, makes this flux divergence free. In the outer collar $\rho = 1$, so the transformed solution and flux agree with the original ones. Smooth boundary voltages have bounded harmonic extensions, so their boundary responses agree; boundedness and density then give equality of the full operators on $H^{1/2}(\partial\Omega)$. The nonzero potential makes the two positive conductivities distinct. Thus the source identity, applied to every bounded harmonic function, is the step from the block's finite family of controlled inputs to the full inverse problem.

Constructing the scalar block. Section 3 proves an exact finite-field scalarization theorem. Under the stated local smoothness and rank conditions, it replaces a uniformly elliptic symmetric tensor by one bounded positive scalar coefficient while preserving the voltage trace and the entire normal-flux functional of each of two designated solutions. Here “finite-field” refers to the number of chosen inputs, not to a finite-dimensional observation of each response.

The product scalar leaves and successive coordinate joins have a classical antecedent in the isotropic approximation of Marino and Spagnolo [8, §3]. The localized flux waves use the polynomial potential method of Raită [11, Theorem 1 and §2]; the proof gives the explicit formula for a surjective symbol and the coupled construction required here. Together with a synchronized coordinate change and a small symmetric tensor correction, these waves produce successive perturbations satisfying the exact equations and preserving both boundary responses. A Baire continuity-point argument then selects a strong limit satisfying one scalar constitutive relation for both fields. This uses the method developed by Kirchheim [7] and adapted to elliptic equations by Astala, Faraco, and Székelyhidi [1, Lemmas 3.7–3.8]. These precedents supply methods; the simultaneous Cauchy-pair conclusion is proved in Section 3.

Section 4 supplies the tensor and the two designated fields. It builds a block with one parent and two child boundary tori, and makes both fields affine in the normal coordinate at all three ends. The construction corrects small residuals across rank-one walls, then transfers a decaying

mode through successively doubled angular frequencies until it vanishes at finite distance. The two resulting current vectors span the zero-sum plane. After scalarization, those two responses and the constant solution cover every separately constant voltage vector on the three boundary components. This is precisely the block property used to force the common surface averages above.

2 Weak equations and changes of variables

We use real functions throughout; complex trace spaces, if desired, follow by complex linear extension. Let $U \subset \mathbb{R}^3$ be bounded and Lipschitz. A tensor is a measurable symmetric matrix A with $aI \leq A \leq bI$ almost everywhere, where $0 < a < b < \infty$. The weak equation is $\operatorname{div}(A\nabla u) = 0$, interpreted as in Equation (1.1). The trace theorem, the zero-trace Poincaré inequality, and coercivity give existence and uniqueness for each trace. A weak solution minimizes its energy among functions of that trace: if $h \in H_0^1(U)$, then

$$\int_U A\nabla(u+h) \cdot \nabla(u+h) = \int_U A\nabla u \cdot \nabla u + \int_U A\nabla h \cdot \nabla h. \quad (2.1)$$

Testing positive and negative truncations gives the usual maximum bound when the trace is bounded. These facts require no derivative of A .

For a divergence-free $F \in L^2(U; \mathbb{R}^3)$, its normal flux is the continuous functional

$$\langle F \cdot \nu, \operatorname{Tr} v \rangle := \int_U F \cdot \nabla v \, dx \quad (v \in H^1(U)). \quad (2.2)$$

It is well defined because the integral vanishes for zero-trace functions. A Cauchy pair consists of $\operatorname{Tr} u$ and this functional for $F = A\nabla u$. This convention also fixes all signs when a function on a block is extended by constants across its boundary components.

We will use that matching traces glue H^1 functions across Lipschitz interfaces. Also $H^1 \cap L^\infty$ is an algebra, with the ordinary weak product rule; Lipschitz compositions on the range of a function obey the weak chain rule. In particular truncation commutes with the trace, so the trace of a bounded H^1 function inherits its bounds. All interfaces used for these operations are finite unions of Lipschitz pieces; infinite constructions are passed to the limit in H^1 explicitly.

For later reference, let $x = X(y)$ be a bi-Lipschitz change of variables that is smooth with nonsingular derivative on open pieces covering almost every point. Write $J = DX$ and set, at $x = X(y)$,

$$\hat{u}(x) = u(y), \quad \hat{A}(x) = \frac{JA(y)J^t}{|\det J|}, \quad \hat{F}(x) = \frac{JF(y)}{|\det J|}. \quad (2.3)$$

Then $\nabla \hat{u} = J^{-t} \nabla u$, and change of variables gives

$$\int_{X(U)} \hat{F} \cdot \nabla \phi \, dx = \int_U F \cdot \nabla(\phi \circ X) \, dy. \quad (2.4)$$

Thus divergence-free fluxes remain divergence free, and the tensor rule preserves the weak energy pairing. A right-hand side transforms as a density: $\hat{r}(X(y)) = r(y)/|\det J(y)|$. Symmetry and uniform ellipticity persist, with bounds depending on the bi-Lipschitz constants. If X is the identity outside an interior box, both Cauchy data are unchanged. We will apply the same rules to potential coordinates and to the piecewise smooth physical collars.

3 Exact scalarization of two Cauchy pairs

This section replaces a matrix conductivity by a scalar one while preserving two specified Cauchy pairs exactly. This is a finite-field statement: it preserves the full flux response to each of two prescribed voltage inputs, not the matrix conductivity's full Dirichlet-to-Neumann operator. Classical isotropic approximation in G -convergence provides useful historical context [8], but no homogenization result is used below. The exact boundary data follow from localized perturbations and a strong limit selected by the Baire theorem. The continuity-point method follows the Baire-category framework for differential inclusions developed by Kirchheim [7] and used for elliptic equations by Astala, Faraco, and Székelyhidi [1, Lemmas 3.7–3.8]. We give the complete localized construction and limiting argument for the simultaneous Cauchy-pair constraints needed here.

Theorem 3.1 (Exact finite-field scalarization). *Let $U \subset \mathbb{R}^3$ be a bounded Lipschitz domain. Let A be a measurable symmetric matrix field satisfying*

$$c_0 I \leq A \leq C_0 I \quad \text{almost everywhere in } U, \quad 0 < c_0 \leq C_0 < \infty.$$

Suppose the real-valued potentials $u_1, u_2 \in H^1(U)$ satisfy $\operatorname{div}(A \nabla u_j) = 0$. Assume that there is an open cover of a full-measure subset of U on whose members A, u_1, u_2 are smooth and have the following local rank property. On each member, a fixed subset of the potentials has everywhere independent gradients, and every remaining potential is a constant affine combination of that subset. The subset may have zero, one, or two elements; a zero-element subset means that both potentials are constant there.

There exist constants $0 < a < b < \infty$, a measurable scalar $a \leq \gamma \leq b$, and $v_1, v_2 \in H^1(U)$ such that

$$\operatorname{div}(\gamma \nabla v_j) = 0, \quad v_j - u_j \in H_0^1(U), \quad (3.1)$$

$$\int_U \gamma \nabla v_j \cdot \nabla \psi \, dx = \int_U A \nabla u_j \cdot \nabla \psi \, dx \quad \text{for every } \psi \in H^1(U). \quad (3.2)$$

The bounds a, b may be chosen using only c_0, C_0 .

We use the Euclidean norm for vectors and the Frobenius norm for matrices. Write Sym_3 for the vector space of real symmetric 3×3 matrices, with this norm. Gradient and flux matrices have their fields as columns. In particular, $E^t F$ is the matrix of their pointwise pairings. The proof will not require a positive lower bound on the independent gradients throughout U : such bounds are used only on individual compactly contained coordinate boxes.

3.1 An open class of finite laminates

For positive triples $\alpha, \beta \in (0, \infty)^3$, a coordinate direction i , and $0 \leq \theta \leq 1$, define their coordinate join $J_i(\theta; \alpha, \beta)$ by

$$[J_i]_i = \left(\frac{\theta}{\alpha_i} + \frac{1-\theta}{\beta_i} \right)^{-1}, \quad [J_i]_\ell = \theta \alpha_\ell + (1-\theta) \beta_\ell \quad (\ell \neq i). \quad (3.3)$$

Fix $0 < a < b$. Let $\mathcal{G}^{\text{diag}}$ consist of the triples obtainable by a finite tree of these joins, starting from scalar triples (s, s, s) with $a < s < b$. A tree is evaluated in one fixed coordinate frame. Let $\mathcal{G}$ be its spectral extension:

$$\mathcal{G} = \{Q \operatorname{diag}(\alpha) Q^t : Q \text{ orthogonal, } \alpha \in \mathcal{G}^{\text{diag}}\}.$$

In particular, every matrix in $\mathcal{G}$ lies strictly between aI and bI .

Lemma 3.2 (Openness and coverage). *The set $\mathcal{G}^{\text{diag}}$ is open in $\mathbb{R}^3$, and $\mathcal{G}$ is open in Sym_3 . Every joining path in a representing tree lies in $\mathcal{G}$. Given c_0, C_0 , the numbers a, b can be chosen so that every symmetric matrix with spectrum in $[c_0, C_0]$ belongs to $\mathcal{G}$.*

Proof. The product construction below is related to the diagonal approximation in Marino and Spagnolo [8, Section 3]; we use its elementary coordinate-join arithmetic directly. First we show that each scalar triple strictly between the endpoints is interior. Given a nearby positive triple $(\alpha_1, \alpha_2, \alpha_3)$, choose $M > \max_i \alpha_i$ and set

$$d_i = \sqrt{1 - \alpha_i/M}.$$

Let t_i independently take the two values $1 - d_i, 1 + d_i$ with equal weights. Its arithmetic mean is one and its harmonic mean is $1 - d_i^2 = \alpha_i/M$. Start with the eight scalar leaves

$$Mt_1 t_2 t_3 I. \tag{3.4}$$

Joining the t_1 pairs in direction 1 gives the diagonal triple

$$(\alpha_1 t_2 t_3, Mt_2 t_3, Mt_2 t_3).$$

Joining the t_2 pairs in direction 2 gives $(\alpha_1 t_3, \alpha_2 t_3, Mt_3)$, and the direction-3 joins give $(\alpha_1, \alpha_2, \alpha_3)$. If the target tends to (s, s, s) , choose M tending to s from above. Every leaf then tends to s , so all leaves lie in (a, b) for sufficiently close targets.

Interior propagates through any nontrivial join. With the other child and a weight $0 < \theta < 1$ fixed, the derivative of the join in its first child is diagonal, with entries θ in the tangential positions and $\theta[J_i]_i^2/\alpha_i^2 > 0$ in the normal position. It is invertible. The inverse function theorem, applied successively up a finite tree, proves openness in diagonal variables. Zero-weight joins can be removed. The diagonal class is permutation invariant; continuity of ordered eigenvalues now proves that its spectral extension is open, including at repeated eigenvalues. Varying the weight of an existing join simply supplies another finite tree with the same children, so its entire path, including its endpoints, stays in $\mathcal{G}$.

For uniform coverage, take $M = 2C_0$ in Equation (3.4). For targets in $[c_0, C_0]^3$, all factors lie between

$$\ell = 1 - \sqrt{1 - c_0/(2C_0)} > 0 \quad \text{and} \quad 2.$$

All leaves therefore lie in $[M\ell^3, 8M]$. For example, $a = M\ell^3/2$ and $b = 16M$ put them strictly inside the leaf interval, uniformly for the entire required spectral range. $\square$

Fix these a, b for the rest of the proof. The distance of a joining path from $\text{Sym}_3 \setminus \mathcal{G}$ can depend on the path, but every fixed path is compact in the open set $\mathcal{G}$ and hence has positive such distance. All corrections below are symmetric and will respect that distance. Thus every corrected matrix remains in the same $\mathcal{G}$, and the global bounds a, b never deteriorate.

We now fix the equations and boundary data that every intermediate field must preserve. Write $A^{\text{in}}, u_1^{\text{in}}, u_2^{\text{in}}$ for the data of Theorem 3.1, and put $F_j^{\text{in}} = A^{\text{in}} \nabla u_j^{\text{in}}$.

Definition 3.3 (Subsolution). A subsolution consists of a measurable symmetric matrix A , two potentials $u_1, u_2 \in H^1(U)$, and the column matrices

$$E = (\nabla u_1, \nabla u_2), \quad F = AE,$$

such that $A \in \mathcal{G}$ almost everywhere and, for $j = 1, 2$,

$$\begin{aligned} u_j - u_j^{\text{in}} &\in H_0^1(U), & \text{div } F_j &= 0, \\ \int_U F_j \cdot \nabla \psi \, dx &= \int_U F_j^{\text{in}} \cdot \nabla \psi \, dx & \text{for every } \psi &\in H^1(U). \end{aligned}$$

It must also have an open cover of a full-measure subset of U on which A, u_1, u_2 are smooth and have the local rank property in Theorem 3.1: on each member one fixed subset of the potentials has independent gradients, and every remaining potential is a constant affine combination of those potentials.

The original fields are a subsolution by Lemma 3.2. The construction will change the matrix and interior fields while keeping the equations and the two entire boundary responses in this definition.

3.2 A split in preliminary coordinates

We first compute the states to be realized by a local perturbation. A divergence-free flux wave with the potentials held fixed can mix the tangential entries of a coordinate join. The normal entry requires its harmonic mean instead. We obtain both at once by coupling the flux wave to a normal change of variables; the following states describe that coupling before localization. Freeze a parent matrix A_0 and its two diagonal children A^+ , A^- in a common orthonormal frame. Let the layering direction be the unit vector d , and let the physical fractions be $\theta_+, \theta_- > 0$, with sum one. A trivial join needs no perturbation. Subscripts d and T denote the normal scalar entry and tangential diagonal block. Set

$$q_s = \frac{A_d^s}{(A_0)_d}, \quad B_d^s = (A_0)_d, \quad B_T^s = q_s A_T^s, \quad \eta_s = \frac{\theta_s}{q_s}, \quad s \in \{+, -\}. \quad (3.5)$$

The normal harmonic-mean formula and the tangential arithmetic-mean formula give

$$\eta_+ + \eta_- = 1, \quad \eta_+ q_+ + \eta_- q_- = 1, \quad \eta_+ B^+ + \eta_- B^- = A_0. \quad (3.6)$$

The fractions η_s , not θ_s , are used before the coordinate change.

Put $\Delta B = B^+ - B^-$ and $\Delta q = q_+ - q_-$. A scalar parameter $z \in [-\eta_+, \eta_-]$ describes the segment

$$B(z) = A_0 + z \Delta B, \quad q(z) = 1 + z \Delta q. \quad (3.7)$$

Its endpoints are the two states in Equation (3.5); its barycenter is $z = 0$. The normal stretch $L(z) = I + (q(z) - 1)d \otimes d$ pushes $B(z)$ forward to

$$C(z) = \frac{L(z)B(z)L(z)^t}{\det L(z)} = \text{diag}(q(z)(A_0)_d, B_T(z)/q(z)) \quad (3.8)$$

in the chosen frame. This entire path is the original coordinate joining path, with changed weights. Indeed if $B = \sum_s \tau_s B^s$ and $q = \sum_s \tau_s q_s$ for $\tau_+ + \tau_- = 1$, the physical weights are $w_s = \tau_s q_s / q$. Their normal harmonic mean is

$$\left(\sum_s \frac{w_s}{A_d^s} \right)^{-1} = q(A_0)_d,$$

and their tangential arithmetic mean is B_T/q . In particular, smoothing and cutting off the scalar parameter while keeping it in its segment will not leave the admissible path.

3.3 Exact localized flux waves

Let $m \in \{1, 2\}$ potentials have independent gradients on a smooth patch. Temporarily discard the affinely dependent columns. For a symmetric matrix to map their gradient matrix E to a flux matrix F , the pairing $E^t F$ must be symmetric. The flux wave must preserve this condition as well as column divergence. After shrinking the patch, choose smooth coordinates

$$z = \Phi(y), \quad z_1 = u_1(y), \dots, z_m = u_m(y),$$

where the independent potentials have been relabeled if needed. If $J = D\Phi$, a flux column becomes the divergence density $JF/|\det J|$. In these coordinates, symmetry of $E^t F$ is exactly symmetry of the leading $m \times m$ block of this transformed flux matrix. Thus the two constraints to preserve are constant coefficient: column divergences vanish and that block is symmetric.

Let V_m be the vector space of $3 \times m$ matrices with symmetric leading block, equipped with the Frobenius inner product. Define

$$D(\xi) : V_m \longrightarrow \mathbb{R}^m, \quad D(\xi)G = \xi^t G.$$

We identify the row vector on the right with a vector in $\mathbb{R}^m$.

The following is the surjective-symbol case of the polynomial potential construction of Raită [11, Theorem 1 and Section 2]. We give the explicit formula used for these coupled constraints.

Lemma 3.4 (A polynomial potential). *For every $\xi \neq 0$, $D(\xi)$ is onto. There is a homogeneous polynomial map $P(\xi) : V_m \rightarrow V_m$ of degree $2m$ such that*

$$D(\xi)P(\xi) = 0, \quad \text{im } P(\xi) = \ker D(\xi) \quad (\xi \neq 0).$$

Consequently every mean-zero smooth periodic plane wave with polarization in $\ker D(\xi)$ admits a compactly supported localization satisfying both constraints exactly, with arbitrarily small absolute uniform norm of the difference from its cut-off leading wave.

Proof. For $m = 1$, surjectivity is ordinary nonzero contraction. For $m = 2$ write

$$G = \begin{pmatrix} a & b \\ b & c \\ d & e \end{pmatrix}, \quad D(\xi)G = (\xi_1 a + \xi_2 b + \xi_3 d, \xi_1 b + \xi_2 c + \xi_3 e).$$

If $\lambda \in \mathbb{R}^2$ annihilates its range, then

$$\xi_1 \lambda_1 = \xi_2 \lambda_2 = \xi_1 \lambda_2 + \xi_2 \lambda_1 = \xi_3 \lambda_1 = \xi_3 \lambda_2 = 0.$$

When $\xi_3 \neq 0$ the assertion follows immediately; otherwise either ξ_1 or ξ_2 is nonzero and the first three equalities give the same conclusion. Thus $D(\xi)D(\xi)^t$ is positive definite at every nonzero ξ , where the transpose uses the specified inner products.

Set $Q(\xi) = D(\xi)D(\xi)^t$ and define

$$P(\xi) = \det Q(\xi)I - D(\xi)^t \text{adj}(Q(\xi))D(\xi). \quad (3.9)$$

The identity $Q \text{adj } Q = (\det Q)I$ gives $DP = 0$ as a polynomial identity. On $\ker D(\xi)$, $P(\xi)$ is multiplication by the nonzero scalar $\det Q(\xi)$, proving the image assertion. Both terms have degree $2m$.

For completeness, let $R \in \ker D(\xi)$ and choose $R_0 \in V_m$ with $P(\xi)R_0 = R$. Let h be a smooth one-periodic function of mean zero and let H be a one-periodic $2m$ -fold primitive, so $H^{(2m)} = h$. Such primitives are obtained by successive integration and subtraction of the mean. Given a smooth compactly supported cutoff χ , apply the constant coefficient differential operator $P(\partial)$ to

$$k^{-2m}\chi(z)H(k\xi \cdot z)R_0. \quad (3.10)$$

Its output lies in V_m and has zero column divergence exactly, by the polynomial identity. Terms with every derivative on H give $\chi(z)h(k\xi \cdot z)R$. Each remaining term has at least one derivative on χ and is $O(k^{-1})$ uniformly for fixed χ, h . This proves the localization claim. $\square$

The flux wave will satisfy the differential constraints exactly but will have a small constitutive error. The following algebraic fact removes that error without losing symmetry.

Lemma 3.5 (Symmetric algebraic correction). *Let E have independent columns and let R satisfy that $E^t R$ is symmetric. With $E^\dagger = (E^t E)^{-1} E^t$, the matrix*

$$H = RE^\dagger + (E^\dagger)^t R^t - (E^\dagger)^t (E^t R) E^\dagger \quad (3.11)$$

is symmetric and satisfies $HE = R$. Its norm is bounded by a locally uniform constant times $\|R\|$ when E ranges over a bounded set with its smallest singular value bounded below.

Proof. Symmetry is immediate. Multiply by E , use $E^\dagger E = I$, and cancel the last two terms using $R^t E = E^t R$. The norm bound follows from the same formula. $\square$

3.4 Realizing one coordinate join

We can now state the local operation used to exhaust a finite laminate tree. Its two conclusions serve different purposes: proximity to the children lowers the remaining tree depth, while weak proximity lets us make that change inside any prescribed weak neighborhood.

Lemma 3.6 (Local realization of a coordinate join). *Let (A, E, F) , with potentials u_1, u_2 , be a subsolution. Let y_0 lie in one of its smooth patches from Definition 3.3, on which one fixed set of $m \in \{1, 2\}$ potentials has independent gradients. Suppose $A_0 = A(y_0)$ is a coordinate join of two children A^+, A^- in one orthonormal frame, with positive weights θ_+, θ_- summing to one and with its joining path contained in $\mathcal{G}$. Given open neighborhoods $\mathcal{O}_+, \mathcal{O}_-$ of the children in Sym_3 and $\delta > 0$, there is an open neighborhood V of y_0 , compactly contained in the patch, with the following property.*

For every open rectangular box Q with $\bar{Q} \subset V$, one can choose a cutoff and periodic waveform, then construct subsolutions (A'_k, E'_k, F'_k) , with potentials $u'_{1,k}, u'_{2,k}$, for all sufficiently large integers k such that $A'_k = A$ outside Q . The differences $u'_{j,k} - u_j$ and $F'_{j,k} - F_j$ vanish outside compact subsets of Q , and

$$\int_U (F'_{j,k} - F_j) \cdot \nabla \psi \, dx = 0 \quad (\psi \in H^1(U), \, j = 1, 2).$$

There are disjoint open subsets $Q_{k,+}, Q_{k,-} \subset Q$ such that

$$A'_k \in \mathcal{O}_s \text{ on } Q_{k,s} \quad (s \in \{+, -\}), \quad |Q \setminus (Q_{k,+} \cup Q_{k,-})| < \delta |Q|.$$

On the same fixed box, with the cutoff and waveform fixed,

$$(E'_k, F'_k) \rightharpoonup (E, F) \quad \text{in } L^2(U) \times L^2(U) \quad \text{as } k \rightarrow \infty.$$

The neighborhood and frequency threshold may depend on all preceding local data, including $\mathcal{O}_+, \mathcal{O}_-$ and δ ; the threshold may also depend on the chosen box, cutoff, and waveform.

Proof. Choose an initial compactly contained neighborhood of y_0 on which the potential chart Φ is defined and the selected gradients have a positive lower singular-value bound. We will choose a smaller neighborhood V from the frozen-error estimates below; the box Q will then have closure in V . Until the constitutive correction is complete, the local matrices E, F, E_0, R and their primed versions contain only these m independent columns. We restore the full pair by the fixed affine relations at the end of the local calculation. Freeze $A_0 = A(y_0)$, $E_0 = E(y_0)$, and $J_0 = D\Phi(y_0)$. For a split as above, the physical polarization $\Delta B E_0$ has zero normal component since the normal row of ΔB vanishes. Its gradient pairing is symmetric. Hence

$$\xi = J_0^{-t} d, \quad R = \frac{J_0 \Delta B E_0}{|\det J_0|} \quad \text{satisfy} \quad R \in V_m, \quad D(\xi)R = 0. \quad (3.12)$$

For the last assertion, contract the displayed matrix with ξ . For the first, the leading rows of J_0 are the gradients of the selected potentials, so their entries are precisely the symmetric pairings with ΔB .

The construction below applies on any box Q with $\overline{Q} \subset V$. After V has been chosen from the estimates below, fix such a box and a smooth cutoff $0 \leq \chi \leq 1$, supported in Q and equal to one except on a thin margin. Choose a smooth periodic function h by convolving the two-state step function that equals η_- on a fraction η_+ of the period and $-\eta_+$ on the remaining fraction η_- . Then

$$\int_0^1 h = 0, \quad -\eta_+ \leq h \leq \eta_-, \quad (3.13)$$

and h equals the two pure values outside an arbitrarily small part of its period. The margin and transition lengths will be chosen to meet the prescribed volume error $\delta |Q|$. Let H_1 be a periodic primitive of h . In Equation (3.10), use the cutoff $\chi \circ \Phi^{-1}$. Transform the resulting flux perturbation back to the original coordinates and denote the perturbed flux by $\widehat{F}_k$. It satisfies

$$\operatorname{div} \widehat{F}_k = 0, \quad E^t \widehat{F}_k = \widehat{F}_k^t E \quad (3.14)$$

exactly, and equals the original flux outside a compact subset of Q . The selected potentials have not yet changed.

Write $\varphi(y) = \xi \cdot \Phi(y)$; thus $\nabla \varphi(y_0) = d$. At the same time define on Q , extending by the identity outside Q ,

$$X_k(y) = y + \frac{\Delta q}{k} d \chi(y) H_1(k\varphi(y)). \quad (3.15)$$

If $z_k(y) = \chi(y)h(k\varphi(y))$, differentiation gives

$$DX_k = I + \Delta q z_k d \otimes \nabla \varphi + O(k^{-1}). \quad (3.16)$$

Throughout Q , this is uniformly close to $L(z_k)$ when V is sufficiently small and then k is large. The scalar z_k stays in the segment in Equation (3.7), because zero lies in that segment and $0 \leq \chi \leq 1$. All $q(z_k)$ are bounded away from zero. More explicitly, the rank-one determinant formula gives

$$\det DX_k = q(z_k) + \Delta q z_k (\partial_d \varphi - 1) + \frac{\Delta q}{k} H_1(k\varphi) \partial_d \chi.$$

First shrink V so that $\partial_d \varphi$ is uniformly close to one; after the box and profiles are fixed, take k large. The displayed determinant, which is the derivative of X_k along each line parallel to d in that direction, is then strictly positive. The displacement is parallel to d and compactly supported. Each such line is consequently mapped monotonically onto itself. This proves that X_k is a global

smooth diffeomorphism, equal to the identity outside Q , and mapping Q onto itself. Its derivative and inverse derivative have bounds independent of sufficiently large k .

Push the potentials and preliminary fluxes forward by this map:

$$u'_{j,k} = u_j \circ X_k^{-1}, \quad E'_k \circ X_k = DX_k^{-t} E, \quad F'_k \circ X_k = \frac{DX_k \hat{F}_k}{\det DX_k}. \quad (3.17)$$

The determinant is positive here. These fields satisfy exact column divergence and pairing symmetry, since

$$(E_k^{t_t} F'_k) \circ X_k = \frac{E^t \hat{F}_k}{\det DX_k}.$$

The selected gradients remain independent. The discarded potentials are recomposed by the same map, so their original constant affine relations with the selected potentials remain valid. Restore their preliminary fluxes by the corresponding constant linear combinations of the selected fluxes, and push them by the same formula. We continue the constitutive estimates on the selected columns.

The errors in the frozen description are only errors in the constitutive approximation, not in these equations. More precisely, let $\omega(V)$ bound the oscillations from their values at y_0 of the fixed smooth fields and chart derivatives on V , with $\omega(V) \rightarrow 0$ as V shrinks to y_0 . The frozen flux error can be seen directly. Put $\Gamma(y) = |\det J(y)| J(y)^{-1}$, so that $\Gamma(y_0)R = \Delta B E_0$. The localized leading wave gives

$$\hat{F}_k - B(z_k)E = (A - A_0)E + z_k[(\Gamma - \Gamma(y_0))R + \Delta B(E_0 - E)] + O(k^{-1}).$$

Consequently, for the fixed plan,

$$\hat{F}_k - B(z_k)E = O(\omega(V)) + O(k^{-1}), \quad (3.18)$$

$$DX_k - L(z_k) = O(\omega(V)) + O(k^{-1}). \quad (3.19)$$

All estimates here are absolute uniform bounds on the box. The constants multiplying $\omega(V)$ depend only on the fixed plan and bounds on the initial neighborhood; z_k remains in its fixed segment. The constants in $O(k^{-1})$ may also depend on the chosen box, cutoff, and waveform. This is why those choices can be fixed before the frequency is increased.

Define the constitutive defect

$$\mathcal{R}_k = F'_k - C(z_k \circ X_k^{-1})E'_k.$$

Writing $K = DX_k$ and evaluating B, L at $z_k(y)$, the pushforward formulas give the exact identity

$$\mathcal{R}_k \circ X_k = \frac{K}{\det K}(\hat{F}_k - BE) + \left(\frac{K B K^t}{\det K} - \frac{L B L^t}{\det L} \right) K^{-t} E.$$

The map bounds and Equations (3.18)–(3.19) therefore give

$$\mathcal{R}_k = O(\omega(V)) + O(k^{-1}). \quad (3.20)$$

Apply Lemma 3.5 to $\mathcal{R}_k$, using the independent columns. Its symmetry condition is exact: both $E_k^{t_t} F'_k$ and $E_k^{t_t} C(z_k \circ X_k^{-1})E'_k$ are symmetric. The selected gradients retain a positive lower singular-value bound, because $E'_k \circ X_k = DX_k^{-t} E$ and the map derivatives are bounded. Thus the correction H_k has a uniform bound by a local constant times $|\mathcal{R}_k|$, and

$$A'_k = C(z_k \circ X_k^{-1}) + H_k$$

maps the selected new gradients to their new fluxes. Restore every discarded flux by the same constant linear combination as its gradient. The common recomposition then gives the full two-column relation $F'_k = A'_k E'_k$ and preserves the original fields outside the box. From this point onward, E, F, E'_k, F'_k and $\widehat{F}_k$ again denote the full two-column matrices; the bound for H_k is the one just obtained from the selected columns.

Choose $\tau > 0$ so that the symmetric τ -neighborhood of the compact path C lies in $\mathcal{G}$ and the τ -balls about A^s lie in $\mathcal{O}_s$ for $s \in \{+, -\}$. This uses distance inside Sym_3 . The coefficient of $\omega(V)$ in the correction bound is fixed on the initial neighborhood, independently of the later box and profiles. Choose V small enough that this contribution is less than $\tau/2$ and that the preceding map determinant is positive after a sufficiently small frequency error. For each box with closure in this V , fix its cutoff and waveform and take k large enough that the remaining contribution is less than $\tau/2$. Then $A'_k \in \mathcal{G}$ throughout the box. The correction needs no bounds on derivatives of the defect. It is smooth inside the box; outside Q keep the original matrix. A jump of the matrix at the box boundary is harmless: the potentials and fluxes agree with the old ones near that boundary, and the matrix is only required to be smooth on an open cover up to null sets.

Where $\chi = 1$ and h is at the pure value for child s , the matrix C is exactly A^s , so the correction bound puts A'_k in $\mathcal{O}_s$. Take $Q_{k,s}$ to be the images under X_k of the open pure regions, omitting the endpoints of the waveform plateaus. The omitted plateau endpoints form a null set. The cutoff margin can have arbitrarily small volume, and the transition intervals can have arbitrarily small total length. The phase is nondegenerate on the box; changing to the chart and integrating in its linear phase direction shows that the preimages of the transition intervals have the corresponding small limiting volume. For the image measure, the determinant formula above gives

$$0 < \det DX_k \leq \sup_{z \in [-\eta_+, \eta_-]} q(z) + |\Delta q| \sup_{z \in [-\eta_+, \eta_-]} |z| \sup_V |\partial_d \varphi - 1| + 1 =: J_*,$$

after the profiles are fixed and k is large enough to make the k^{-1} term in the determinant formula at most one in absolute value. Thus J_* is fixed by the preceding local data and V , independently of the cutoff margin and transition lengths, and bounds the volume increase under X_k . Choose the margin and transition lengths so that, for all sufficiently large k , the complement of $Q_{k,+} \cup Q_{k,-}$ has volume less than $\delta|Q|$. The new smooth box pieces and the old pieces away from its faces retain the full-measure local rank cover.

Boundary data and weak proximity. Every perturbation just constructed preserves the outer boundary traces and fluxes exactly. Before the pushforward the flux change is divergence-free and compactly supported in the interior. After the pushforward the same is true of its difference from the old flux, since X_k is the identity outside Q . Thus for every $\psi \in H^1(U)$ the flux difference has zero pairing with $\nabla\psi$. To see this without any boundary smoothness of the fields, insert a smooth cutoff equal to one on the support of the flux difference, and use density of smooth tests in $H_0^1(U)$ together with its distributional zero divergence. The potentials also agree outside a compact subset of Q , so their trace difference vanishes. There is no new global boundary compatibility condition: the perturbation starts from an existing solution pair and satisfies the exact local equations with compact support. Together with the constitutive relation and the local rank cover, these facts prove that the corrected fields are subsolutions.

It is equally important that these perturbations can be made arbitrarily weakly close to the old fields, even though the local child states stay separated. Keep the chosen V , box, waveform, and cutoff fixed. The preliminary flux perturbation is a smooth matrix amplitude times $h(k\varphi(y))$, plus a uniformly $O(k^{-1})$ error. Changing to the chart and using the zero mean of h shows

$$\widehat{F}_k \rightharpoonup F \quad \text{in } L^2. \tag{3.21}$$

For example, for a smooth compactly supported test amplitude one can integrate a bounded periodic primitive in the phase direction, obtaining $O(k^{-1})$; density and the uniform L^2 bound give the stated weak convergence.

The product of the oscillating Jacobian and flux in Equation (3.17) cannot be treated by multiplying weak limits. Instead write $v_k = X_k - I$. For a smooth vector test function ψ , change variables to obtain

$$\int_U F'_k(x) \cdot \psi(x) \, dx = \int_U DX_k(y) \hat{F}_k(y) \cdot \psi(X_k(y)) \, dy$$

for each flux column. Replacing $\psi \circ X_k$ by ψ costs $o(1)$ because $v_k \rightarrow 0$ uniformly and the other factors are uniformly bounded in L^2 . The remaining additional term vanishes by the exact divergence equation:

$$\int_U \partial_j (v_k)_i (\hat{F}_k)_j \psi_i \, dy = - \int_U (v_k)_i (\hat{F}_k)_j \partial_j \psi_i \, dy \rightarrow 0. \quad (3.22)$$

Summation over repeated spatial indices is understood only in this display. Equations (3.21) and (3.22) prove $F'_k \rightharpoonup F$. The potentials converge uniformly on the modified compact region, since $X_k^{-1} \rightarrow I$ uniformly and the old potentials are smooth there. Their gradients are uniformly bounded in L^2 , either directly on the box or by the energy estimate below. Consequently $E'_k \rightharpoonup E$ as well. The same weak limits hold for the restored columns, since their local changes are the same fixed linear combinations of the selected-column changes.

The choices now have the order asserted in the lemma. First the neighborhood V controls the fixed coefficient error $O(\omega(V))$ for every box whose closure lies in V . After fixing one such box, choose its cutoff and waveform to meet the volume budget. All sufficiently large frequencies then give the pure-state membership, and that same family converges weakly to the old pair as the frequency tends to infinity. The freezing error is not an error in the differential constraints and does not obstruct this limit. $\square$

3.5 Finite-depth exhaustion

We use Lemma 3.6 to replace a finite laminate tree by its children one depth at a time. Every step remains a subsolution in the explicit sense of Definition 3.3.

Proposition 3.7 (Weak approximation by scalar-near matrices). *Given a subsolution, a weak L^2 neighborhood of its pair (E, F) , and $\varepsilon > 0$, there is a subsolution in that neighborhood whose matrix has distance less than ε from the scalar matrices outside a set of volume less than ε .*

Proof. For clarity we make the finite-depth argument explicit. In diagonal variables let T_0 be the triples in $\mathcal{G}^{\text{diag}}$ whose distance from scalar triples is strictly less than ε . Inductively, T_k contains T_{k-1} and the outputs of joins of two T_{k-1} children whose entire joining path lies in $\mathcal{G}^{\text{diag}}$. Equivalently these are targets with a representing tree of depth at most k , ending in T_0 , with the indicated path condition. Take their spectral extensions when discussing matrices.

Each T_k is open and permutation invariant. This is immediate for T_0 . For a nontrivial join with children in T_{k-1} , its invertible derivative in one child, computed in Lemma 3.2, absorbs any sufficiently small change of output into a change of that child. The child remains in the open set T_{k-1} , and the whole path remains in the open class by compactness. This proves the induction; permutation invariance and spectral openness follow as before. Every member of $\mathcal{G}$ lies in some T_k , since its defining tree ends at scalar leaves.

Intersect the smooth rank patches with the set where $A \in \mathcal{G}$. This intersection is open in each patch and still has full measure. Its subsets where $A \in T_k$ are open and increase to the

whole intersection. By finite measure, choose a finite K so that the portion not covered by these rank-regular T_K patches has volume less than $\varepsilon/3$. If $K = 0$, the original subsolution already suffices. Otherwise fix $\delta > 0$ with $\delta|U| < \varepsilon/(3K)$. At each of the K reductions, take a compact subset of the current good region losing volume less than $\varepsilon/(3K)$, and use this fixed relative tolerance δ in the local lemma before choosing its neighborhoods. Consider one reduction from T_k to T_{k-1} , with $1 \leq k \leq K$. Choose every admissible neighborhood below inside its current open T_k rank patch.

At a rank-positive point, fix its smooth potential chart. If its matrix already belongs to T_{k-1} , choose an admissible neighborhood on which $A \in T_{k-1}$; every contained box then needs no change. Otherwise choose its frozen two-child plan, with children in T_{k-1} . Apply Lemma 3.6 with both child neighborhoods equal to the spectral extension of T_{k-1} and with the already fixed relative tolerance δ . It supplies an admissible neighborhood for every contained grid box; the large-frequency family then permits any prescribed weak error. The local independent gradients have a positive smallest singular value on the initial compact neighborhood used by that lemma. At a rank-zero point, choose an admissible neighborhood on which the potentials are constant and their fluxes vanish. On any box with closure inside it, replace the matrix by any scalar strictly in (a, b) , without changing the fields.

These choices give admissible neighborhoods of every point in the chosen compact subset. A finite subcover of that compact set and a fine enough grid give finitely many disjoint box interiors, with closures inside admissible neighborhoods, covering the compact set except for grid faces. The frozen point need not lie in the grid box: Lemma 3.6 applies to every box whose closure lies in its admissible neighborhood. One can equivalently refine a grid successively. Each box is treated using its own chart, fixed frame, rank bound, and frozen plan; no continuous eigenframe on the whole domain is required.

For this reduction, construct every boxwise output from the same incoming subsolution, with the frozen data just selected. In each rank-positive box requiring a join, use the local lemma with the fixed relative tolerance δ . Paste the individual potential and flux differences and the matrix choices over the disjoint box interiors. The potential and flux differences have compact support in those boxes, so the pasted gradients and fluxes have each box's constitutive relation, and the exact flux pairings add. Since the box interiors are disjoint, the sum of the exceptional volumes is less than $\delta \sum_Q |Q| \leq \delta|U| < \varepsilon/(3K)$. The unchanged T_{k-1} boxes and the rank-zero scalar boxes create no exceptional volume. Together with them, the open pure regions give smooth patches with the local rank property on which the new matrix belongs to T_{k-1} . The output is again a subsolution. In joined boxes this is part of the lemma; in the other boxes the fields are unchanged, and either the matrix is unchanged or it maps the zero gradients to the zero fluxes. The scalar replacement lies in $\mathcal{G}$ and retains the local rank property. Matrix jumps at the finitely many box faces add only null seams, and the original cover remains where nothing changed.

Perform the next step only inside these new good patches and continue down to T_0 . Untreated exceptional portions can be left untouched. Every map is supported in a selected box and maps that box onto itself. Thus later operations can be confined to the current good set. The initial loss is less than $\varepsilon/3$; the K compact-subset losses and the K local exceptional losses are each less than $\varepsilon/(3K)$ per stage. Their total is less than ε . Fixed global rank bounds or fixed tolerances over infinitely many plans are not needed.

Finally, because the box outputs are pasted from one incoming pair, the finite sum of their field differences is arbitrarily weakly small by taking their frequencies sufficiently large. Use a metric for the weak topology on the bounded set of all subsolution pairs, whose boundedness is shown below, and allot a sufficiently small part of the prescribed neighborhood to each of the finitely many stages. The final pair lies in the prescribed neighborhood, and its matrix is in T_0 off a set of measure less than ε , as asserted. $\square$

3.6 The strong limit selected by Baire

We now close the argument without losing either Cauchy condition. All subsolution pairs are uniformly bounded in

$$\mathcal{H} = L^2(U; \mathbb{R}^{3 \times 2}) \times L^2(U; \mathbb{R}^{3 \times 2}).$$

Indeed, for each potential $\tilde{u}_j$ of a subsolution, energy minimality with its fixed boundary trace gives

$$a \int_U |\nabla \tilde{u}_j|^2 \leq \int_U \tilde{A} \nabla \tilde{u}_j \cdot \nabla \tilde{u}_j \leq b \int_U |\nabla u_j^{\text{in}}|^2. \quad (3.23)$$

Here $\tilde{A}$ is that subsolution's matrix. Also $|F| \leq b|E|$. Since $\tilde{u}_j - u_j^{\text{in}} \in H_0^1(U)$, the zero-trace Poincaré inequality bounds the potentials themselves as well.

Let Z be the weak closure in $\mathcal{H}$ of all subsolution pairs. It is nonempty, weakly compact, and metrizable in its weak topology. One may use a metric obtained from a countable orthonormal basis of the separable Hilbert space $\mathcal{H}$. Every point of Z still consists of gradients of potentials with the prescribed traces and of divergence-free fluxes with the prescribed normal-flux functionals. To verify the gradient assertion, take a weakly convergent sequence of subsolutions; Equation (3.23) and Poincaré give weakly convergent potentials in the fixed affine space $u_j^{\text{in}} + H_0^1(U)$. Their weak gradients are the limiting columns. For the flux assertion, all functionals $F_j \mapsto \int_U F_j \cdot \nabla \psi$ are weakly continuous, for every fixed $\psi \in H^1(U)$.

Lemma 3.8 (Weak-to-strong continuity points). *The identity from Z with its weak topology to $\mathcal{H}$ with its norm topology has a dense set of continuity points.*

Proof. Let P_N be the finite-rank orthogonal projections onto the first N vectors of a fixed orthonormal basis. Their restrictions to Z are weak-to-norm continuous and $P_N z \rightarrow z$ in norm for every z . Here is the relevant Baire argument explicitly. On any nonempty closed subset K of Z and for $\delta > 0$, the closed sets

$$K_N = \{z \in K : \|P_p z - P_q z\| \leq \delta \text{ for all } p, q \geq N\}$$

cover K . Compact metric spaces are Baire, so some K_N has nonempty relative interior. On that patch, letting p tend to infinity gives $\|z - P_N z\| \leq \delta$. Shrinking the patch using continuity of P_N , its strong diameter is at most 3δ .

Apply this inside the closure of an open set whose closure lies in any prescribed nonempty open subset of Z . The relative patch meets the original open set, because that set is dense in its closure. Their intersection contains a nonempty open subset of Z of strong diameter at most 3δ . Consequently, for each $r > 0$, the union of all open subsets of Z of strong diameter less than r is open and dense. The intersection of these sets for $r = 1, 1/2, 1/3, \dots$ is dense by Baire, and every point of the intersection is a continuity point of the identity. $\square$

Proof of Theorem 3.1. Choose a continuity point $(E, F) \in Z$ from Lemma 3.8. By the definition of Z , choose subsolution pairs converging weakly to it, and apply Proposition 3.7 to each, with weak errors tending to zero and with $\varepsilon_j = 2^{-j}$. Denote the resulting subsolutions by (A_j, E_j, F_j) . Then

$$(E_j, F_j) \longrightarrow (E, F) \quad \text{strongly in } \mathcal{H}, \quad (3.24)$$

and $\text{dist}(A_j, \mathbb{R}I) < 2^{-j}$ except on a set of measure less than 2^{-j} . Put $\lambda_j = \text{tr}(A_j)/3$. Since $A_j \in \mathcal{G}$, one has $a < \lambda_j < b$, and the Frobenius-distance projection onto scalar matrices is exactly $\lambda_j I$.

The exceptional measures are summable. After discarding their limsup, which has measure zero, we have $A_j - \lambda_j I \rightarrow 0$ pointwise. Passing to a subsequence in Equation (3.24) also gives pointwise convergence of E_j, F_j almost everywhere. At such a point,

$$F_j - \lambda_j E_j = (A_j - \lambda_j I)E_j \longrightarrow 0.$$

If $E \neq 0$, it follows that

$$\lambda_j \longrightarrow \gamma = \frac{E : F}{|E|^2} \in [a, b], \quad F = \gamma E, \quad (3.25)$$

where $E : F$ is the Frobenius pairing. If $E = 0$, the bound $|F_j| \leq b|E_j|$ forces $F = 0$; define $\gamma = a$ there. This defines one measurable scalar, common to both columns, with $a \leq \gamma \leq b$ almost everywhere. No pointwise subsequence of the bounded coefficient sequence itself was assumed.

The strong convergence is essential for excluding concentration on small exceptional sets. Equivalently, it makes $|E_j|^2$ uniformly integrable, so the coefficient defect times E_j also tends to zero in L^2 . A mere uniform energy bound would not suffice for that conclusion.

Finally the gradient and flux characterization of Z supplies $v_1, v_2 \in H^1(U)$ with the original traces, gradients E , and the original normal-flux functionals. Since $F = \gamma E$ and its columns have zero divergence, these are precisely Equations (3.1) and (3.2). $\square$

4 Constructing the branching block

We now construct a building block with three boundary components. Its essential property is that separately constant voltages produce constant flux densities in prescribed torus coordinates. All constants in this section may depend on this one fixed block.

Proposition 4.1 (Branching block). *Let $B \subset \mathbb{R}^3$ be a bounded connected Lipschitz domain whose boundary has three components $\Gamma_0, \Gamma_1, \Gamma_2$. Suppose these components have disjoint product collars parametrized by $\mathbb{T}^2 \times [0, \eta_i]$, where $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$. Assume the parametrizations are bi-Lipschitz, smooth with nonsingular derivative on open pieces covering up to a null set, and that removing sufficiently thin collars leaves a connected Lipschitz central region. Let μ_i be the probability measure obtained by pushing forward $(2\pi)^{-2} d\theta$ to Γ_i by its collar parametrization.*

There is a scalar conductivity γ_B , bounded above and bounded away from zero, with the following property. For every $c = (c_0, c_1, c_2) \in \mathbb{R}^3$, the weak γ_B -harmonic function U_c with boundary value c_i on Γ_i satisfies

$$\int_B \gamma_B \nabla U_c \cdot \nabla \psi \, dx = \sum_{i=0}^2 I_i(c) \int_{\Gamma_i} \psi \, d\mu_i, \quad \psi \in H^1(B), \quad (4.1)$$

where $\sum_i I_i(c) = 0$. The linear map $c \mapsto I(c)$ has kernel $\mathbb{R}(1, 1, 1)$ and image $\{I \in \mathbb{R}^3 : \sum_i I_i = 0\}$.

We first construct a uniformly elliptic symmetric tensor and two fields with the required Cauchy data. Only at the end will we apply Theorem 3.1.

4.1 Attaching flat ends

Write B_0 for a connected Lipschitz central region after removing collars. Attach an auxiliary cylinder $\mathbb{T}^2 \times [0, \infty)$ to each component of ∂B_0 , using the corresponding angular coordinates. The coordinate t increases away from B_0 . These infinite cylinders are auxiliary: we will retain only finite portions of them.

On B_0 use the identity tensor. On each cylinder use the energy

$$\int_0^\infty \int_{\mathbb{T}^2} (|\partial_t p|^2 + \nabla_\theta p \cdot S \nabla_\theta p) \, d\theta \, dt, \quad (4.2)$$

where S is a fixed positive definite symmetric 2×2 matrix such that the numbers $h^t S h$, $h \in \mathbb{Z}^2 \setminus \{0\}$, have no equalities except those between h and $-h$. Such an S exists: each unwanted equality cuts

out a proper hyperplane in the space of symmetric matrices, and countably many proper hyperplanes do not cover the open positive cone. The hyperplane is proper because $hh^t = ll^t$ implies $l = \pm h$.

Set $\lambda_h = (h^t S h)^{1/2}$. For a trace f on a torus, let $\bar{f} = (2\pi)^{-2} \int f$ and write $f = \bar{f} + \sum_{h \neq 0} \hat{f}(h) e^{ih \cdot \theta}$. Ellipticity gives $\lambda_h \geq c_S |h|$ for some $c_S > 0$, so only finitely many frequencies have rate below any fixed bound. Given a slope s , its continuation into the end is

$$p(t, \theta) = \bar{f} + st + \sum_{h \neq 0} \hat{f}(h) e^{-\lambda_h t} e^{ih \cdot \theta}. \quad (4.3)$$

Only the decaying part has finite energy on the infinite end; the full field is locally H^1 there. Its derivative at $t = 0$, directed into the end, is $s - Pf$, where P is the Fourier multiplier λ_h on nonconstant modes and zero on constants.

For prescribed slopes $s_0 + s_1 + s_2 = 0$, solve on B_0 , modulo a common constant,

$$\int_{B_0} \nabla p \cdot \nabla \psi \, dx + \sum_{i=0}^2 \langle P p_i, \psi_i \rangle = \sum_{i=0}^2 s_i \int_{\mathbb{T}^2} \psi_i \, d\theta. \quad (4.4)$$

Here $p_i, \psi_i \in H^{1/2}(\mathbb{T}^2)$ are traces in the attachment coordinates. The brackets denote the $H^{-1/2}(\mathbb{T}^2), H^{1/2}(\mathbb{T}^2)$ duality, equivalently

$$\langle Pf, g \rangle = (2\pi)^2 \sum_{h \neq 0} \lambda_h \hat{f}(h) \hat{g}(-h).$$

The trace forms on the left are continuous on $H^1(B_0)$. For completeness, in a short collar the estimate for a Fourier mode with rate $m > 0$ is

$$m |q(0)|^2 \leq C \int_0^\eta (|q'|^2 + (1 + m^2) |q|^2) \, dt.$$

It follows by applying the fundamental theorem of calculus and Cauchy–Schwarz on an interval of length comparable to $\min\{\eta, m^{-1}\}$; summation over modes and the bi-Lipschitz collar bounds give the assertion. The central Dirichlet energy is coercive modulo constants by the Poincaré inequality on connected B_0 ; the added trace forms are nonnegative. The right side kills constants. Hilbert space minimization therefore gives Equation (4.4). Together with Equation (4.3), it gives a joint weak solution, since the central outward flux functional equals $s_i - P p_i$.

Choose two fields corresponding to a basis of $\{s \in \mathbb{R}^3 : \sum_i s_i = 0\}$. On each end the slope functional on this two-dimensional space is nonzero. Make an invertible affine change of these two fields, on this end only, so that the new fields u, v have respective slopes 1, 0 and both have zero constant offset. At the end of our modification we will undo this same affine change. Consequently the original global pair and its slope vectors will be preserved.

If v is not zero on the end, its first nonzero Fourier eigenspace consists of a single pair of opposite frequencies. A unimodular change of torus coordinates, followed by a translation, puts that leading term in the form

$$b e^{-\lambda t} \cos(kx), \quad b \neq 0, \quad k \in \mathbb{N}, \quad \lambda > 0. \quad (4.5)$$

Indeed, divide the frequency vector by the greatest common divisor of its coordinates and complete the resulting primitive vector to an integral unimodular basis. Translation absorbs the phase. The angular tensor remains constant and positive definite under this change. Both changes preserve uniform probability measure on the torus, so they do not alter the measures prescribed in Proposition 4.1.

4.2 Matching one flat end exactly

The next lemma replaces the asymptotic form of the two fields by exact data on a finite section of the end. This leaves a single mode for the frequency construction in the following subsection.

Lemma 4.2 (Exact matching on one flat end). *Fix one end after the preceding affine and angular normalizations, and write $A_0 = \text{diag}(1, S)$ in its coordinates (t, x, y) . Let $u_{\text{in}}, v_{\text{in}}$ be the two flat continuations in Equation (4.3), with zero constant offsets and respective slopes 1, 0. Suppose first that v_{in} is nonzero, with leading term $be^{-\lambda t} \cos(kx)$ as in Equation (4.5); thus $\lambda^2 = k^2 S_{xx}$.*

For every sufficiently large T , there are smooth fields u_T, v_T on $\mathbb{T}^2 \times [T, T+4]$ and a smooth symmetric matrix H_T compactly supported in the interior of this band such that

$$\begin{aligned} \text{div}((A_0 + H_T)\nabla u_T) &= \text{div}((A_0 + H_T)\nabla v_T) = 0, \\ \|H_T\|_{L^\infty} &\longrightarrow 0 \quad (T \rightarrow \infty). \end{aligned} \tag{4.6}$$

Near the left edge the pair is exactly $(u_{\text{in}}, v_{\text{in}})$; near the right edge it is exactly $(t, be^{-\lambda t} \cos(kx))$. The fluxes in the positive t direction there are respectively $(\partial_t u_{\text{in}}, \partial_t v_{\text{in}})$ and $(1, -\lambda be^{-\lambda t} \cos(kx))$. For large T , the corrected tensor $A_0 + H_T$ is uniformly elliptic. The construction also has $\partial_t u_T > 0$, and the two gradients are independent off a union of $2k$ smooth walls. If v_{in} is identically zero, one instead obtains $v_T = 0$, right-hand field and flux pairs $(t, 0)$ and $(1, 0)$, and all the stated conclusions for H_T, u_T .

The gradients of the target pair become dependent on the $2k$ walls $\sin(kx) = 0$. A correction based only on the potential coordinates (u, v) therefore cannot cover the band. The proof will remove the source near each wall, use wall fluxes to balance the remaining source between the regular strips, and then solve separately in those strips. We first record the compatibility conditions and the regular solve.

All changes of coordinates in this argument are made in the energy form. Thus matrices, their flux columns, and sources in a new chart are the corresponding matrix, vector, and source *densities*. The divergence equations, symmetry of a matrix density, and integrals of source densities are preserved by these changes of coordinates.

For a compactly supported symmetric correction H , the two sources $\text{div}(H\nabla u)$ and $\text{div}(H\nabla v)$ have zero integral. Symmetry also gives

$$v \text{div}(H\nabla u) - u \text{div}(H\nabla v) = \text{div}(vH\nabla u - uH\nabla v), \tag{4.7}$$

so their cross-moment has zero integral as well. On a connected region where the gradients are independent, these conditions are sufficient for data of fixed compact support, as the following lemma shows.

First recall an elementary compact divergence construction. On a rectangular box Q , every $r \in C_c^\infty(Q)$ with $\int_Q r = 0$ is the divergence of a vector in $C_c^\infty(Q; \mathbb{R}^d)$. To see this inductively, write $Q = (a, b) \times Q'$, choose $\eta \in C_c^\infty(a, b)$ with $\int \eta = 1$, and put $R(x') = \int_a^b r(t, x') dt$. The first component

$$F_1(x) = \int_a^{x_1} (r(t, x') - \eta(t)R(x')) dt$$

is compactly supported, and the remaining source is $\eta(x_1)R(x')$. Apply the construction in Q' and multiply its components by η . The one-dimensional case is direct integration. With fixed boxes and fixed auxiliary bumps, this is a linear construction with bounds in each smooth norm. The same assertion holds for data supported in a fixed compact subset of a connected open region: use a finite connected cover by boxes, a partition of unity, and bumps in overlaps to transfer the mean of each piece along a spanning tree. The final piece has zero mean because the original data do.

Lemma 4.3 (Two symmetric divergence columns). *Suppose u, v are smooth with independent gradients on a connected open region. Let r^1, r^2 be smooth densities supported in a fixed compact subset of that region and satisfying*

$$\int r^1 = \int r^2 = \int (vr^1 - ur^2) = 0. \quad (4.8)$$

Then there is a compactly supported smooth symmetric matrix density H such that

$$\operatorname{div}(H\nabla u) = r^1, \quad \operatorname{div}(H\nabla v) = r^2. \quad (4.9)$$

For a fixed finite system of regular coordinate charts and supports, the construction has bounds in smooth norms. These bounds persist under sufficiently small smooth perturbations of the charts and fields.

Proof. Use local coordinates $y_1 = u, y_2 = v, y_3$ and cover the source support by finitely many such coordinate boxes joined by overlaps in the regular region. A partition of unity splits the data. In any nonempty open overlap, smooth test pairs generate arbitrary vectors of the three moments in Equation (4.8). For example, take two bumps in the first component with different averages of v , and one bump in the second component. Their moment vectors are $(1, 0, \bar{v}_1)$, $(1, 0, \bar{v}_2)$, and $(0, 1, -\bar{u}_3)$, which are independent when $\bar{v}_1 \neq \bar{v}_2$. Such bumps exist because v is a coordinate. Transfer the moment vector of each piece along a spanning tree to one final box; the total vector there is zero. We have reduced to a pair satisfying all three conditions in a single box.

Solve $\operatorname{div} F^i = r^i$ compactly by the box construction. Put $q = F_2^1 - F_1^2$. Integration by parts gives

$$\int (y_2 r^1 - y_1 r^2) = - \int q = 0.$$

Choose a compactly supported vector h with $\operatorname{div} h = q$. Add to the first column the vector

$$(\partial_2 h_1, -\partial_1 h_1 - \partial_3 h_3, \partial_2 h_3)^t$$

and to the second column the vector

$$(\partial_2 h_2, -\partial_1 h_2, 0)^t.$$

Both additions have zero divergence, and their change to $F_2^1 - F_1^2$ is $-\operatorname{div} h = -q$. The corrected columns can therefore be completed to a symmetric matrix: set its third-row counterparts by symmetry and its unused 33 entry to zero. Since $\nabla y_1 = e_1$ and $\nabla y_2 = e_2$, its first two columns give Equation (4.9). Transform back to the original coordinates.

All operations involve fixed compact supports, fixed integrations, smooth multipliers, and finitely many derivatives. The overlap moment matrices have nonzero determinants; they and the coordinate Jacobians remain invertible under small smooth perturbations. This proves the stated uniform bounds, allowing a fixed finite loss of derivatives. $\square$

Proof of Lemma 4.2. We first treat the nonzero case. Write the translated band as $0 < t < 4$, and normalize the input fields there by

$$\tilde{u}_T(t, \theta) = u_{\text{in}}(T + t, \theta) - T, \quad \tilde{v}_T(t, \theta) = \frac{v_{\text{in}}(T + t, \theta)}{be^{-\lambda T}}.$$

They tend in every fixed smooth norm on the band to

$$U = t, \quad V = e^{-\lambda t} \cos(kx). \quad (4.10)$$

Indeed, the rate spectrum is discrete, so the first omitted rate in v_{in} , if present, is strictly larger than λ ; if there is no omitted mode, the normalized second field is already V . For an $H^{1/2}$ trace, Cauchy–Schwarz bounds every differentiated Fourier series on a translated band by the trace norm times a convergent sum of polynomial weights against the exponential factors. The positive rate gap then controls the normalized remainder; the decaying part of $\tilde{u}_T$ is handled in the same way.

Choose a fixed smooth cutoff which is zero near $t = 0$ and one near $t = 4$. Interpolate from $(\tilde{u}_T, \tilde{v}_T)$ to (U, V) with this cutoff, and call the resulting pair u, v . Both endpoint pairs are A_0 -harmonic. Hence the source densities $r^1 = -\text{div}(A_0 \nabla u)$, $r^2 = -\text{div}(A_0 \nabla v)$ for the flat tensor are supported in a fixed interior subband and tend to zero in every fixed smooth norm.

These sources satisfy Equation (4.8) exactly. The first two equalities express equality of the mean axial slopes before and after interpolation. For the third, integrate the counterpart of Equation (4.7) for A_0 . The boundary expression is the difference of the cross-flux concomitants

$$\int_{\mathbb{T}^2} (v \partial_t u - u \partial_t v) \, d\theta.$$

For the normalized input pair, continued harmonically along the translated end, this is independent of t and tends to zero at infinity: its first field is linear plus decaying modes, and its second has zero offset and zero slope. For the pure pair it vanishes by angular integration. Hence the difference is zero.

The three global moments now vanish, but the gradients may still fail to be independent near the walls of the limiting pair. For sufficiently large T , use coordinates $(s = u, x, y)$, since u_t is close to one. Here v_x denotes differentiation with s, y held fixed. Near each limiting wall, the equation $v_x = 0$ has a unique nearby smooth graph $x = x_j(s, y)$ by the implicit function theorem. Recenter by $z = x - x_j(s, y)$. Then

$$v_z(s, 0, y) = 0, \quad |v_{zz}(s, 0, y)| \geq c_* > 0 \quad (4.11)$$

on a fixed narrow strip. After narrowing it if necessary, v_z/z extends smoothly and is bounded away from zero on the strip. The graphs and coordinates are periodic in y and converge smoothly to their limiting counterparts. Choose all $2k$ strips disjoint.

We now remove the source near each wall. In its chart, r^1, r^2 and H denote the transformed densities under the convention above, and derivatives hold the other recentered coordinates fixed. Set

$$\begin{aligned} \alpha(s, z, y) &= \int_0^z r^1(s, q, y) \, dq, \\ G(s, z, y) &= \int_0^z (r^2 - \partial_s(\alpha v_z))(s, q, y) \, dq - \alpha(s, z, y) v_s(s, z, y). \end{aligned} \quad (4.12)$$

Use only the s, z block of H , taking

$$H_{ss} = 0, \quad H_{sz} = H_{zs} = \alpha, \quad H_{zz} = G/v_z.$$

All entries involving y are zero. The quotient extends smoothly across $z = 0$: the numerator vanishes there, so G/z is smooth, while v_z/z is smooth and bounded away from zero. In particular

$$H_{zz}(s, 0, y) = \frac{r^2(s, 0, y) - r^1(s, 0, y) v_s(s, 0, y)}{v_{zz}(s, 0, y)}.$$

Differentiating Equation (4.12) gives

$$\partial_z \alpha = r^1, \quad \partial_z G = r^2 - \partial_s(\alpha v_z) - \partial_z(\alpha v_s).$$

Since $u = s$, the two flux columns are $(0, \alpha, 0)^t$ and $(\alpha v_z, \alpha v_s + G, 0)^t$; the displayed identities prove both equations in Equation (4.9). The fixed denominator bound and the smooth convergence of the wall charts show that this correction is small in every fixed smooth norm. Integration only in z preserves its support away from the axial ends. Multiply it by a cutoff in z which equals one in a narrower wall strip and vanishes outside the chosen strip. After subtracting the divergences of these compact corrections, the remaining sources are supported away from all walls. They still have three zero total moments by Equation (4.7).

The remaining source pair may nevertheless have nonzero moments in an individual regular strip between successive walls. We next transfer these imbalances across the walls without recreating a source there. To do so, choose any smooth $a(s, y)$ supported in a fixed middle axial interval, put $H_{sz} = H_{zs} = a$, $H_{ss} = 0$, and set

$$H_{zz}v_z = -2a(v_s(s, z, y) - v_s(s, 0, y)) - a_s(v(s, z, y) - v(s, 0, y)). \quad (4.13)$$

The first divergence is zero because a is independent of z . For the second, the z derivative of the right side is $-2av_{sz} - a_s v_z$, which cancels $\partial_z(av_s) + \partial_s(av_z)$. Differentiating $v_z(s, 0, y) = 0$ with the recentered z fixed gives $v_{sz}(s, 0, y) = 0$. Hence both differences on the right of Equation (4.13) are of order z^2 . Thus division by v_z is again smooth and $H_{zz}v_z$ vanishes on the wall. Cut this matrix off away from the wall. Its new sources are confined to the adjacent regular strips. Their three moments on one side, up to a common orientation sign, are

$$\int a(s, y)(1, v_s, v - sv_s)(s, 0, y) \, ds \, dy; \quad (4.14)$$

on the other side they are their negatives. This follows by integrating the two fluxes and Equation (4.7); their normal values on the wall are a , av_s , and $a(v - sv_s)$. They are flux densities in this chart, so the wall integral uses $ds \, dy$.

The map in Equation (4.14) has a bounded right inverse uniformly for large T . On a limiting wall write $V(s, 0, y) = \sigma e^{-\lambda s}$, where $\sigma \in \{1, -1\}$, and set

$$w(s) = (1, -\sigma \lambda e^{-\lambda s}, \sigma(1 + \lambda s)e^{-\lambda s}).$$

Its three component functions are linearly independent on every open interval: after multiplying a linear relation by $e^{\lambda s}$, two derivatives force its exponential coefficient to vanish, and the remaining affine coefficients then vanish. Choose a nonnegative smooth middle-supported function χ , positive on an interval, and use $a_j(s, y) = \chi(s)w_j(s)$, $j = 1, 2, 3$. The limiting moment matrix is $2\pi \int \chi(s)w(s)w(s)^t \, ds$, which is positive definite. Its determinant remains nonzero for the smoothly perturbed wall coordinates and fields. This provides a fixed finite-dimensional right inverse with the asserted support and bounds.

The regular strips form a cyclic adjacency graph with $2k$ vertices. Choose a spanning tree and transfer each strip's three-vector imbalance towards a root. The root imbalance is zero because the total is zero. There are only finitely many transfers, so all corrections remain small with the sources. Their cutoff regions lie in fixed cores of the regular strips. Each regular strip is connected; away from the narrow wall neighborhoods v_x has a fixed nonzero sign and (u, v, y) are regular local coordinates. The remaining source pair in each strip now has all three moments zero and support in a fixed compact subset of that strip. Lemma 4.3 removes it there.

Choose the wall strips and their transverse cutoffs first for the limiting pair in Equation (4.10), with the axial supports a positive distance from the band ends. In the complementary regular strips choose the source cores a positive distance from their bounding walls and the band ends, and fix the associated finite chart covers, overlap bumps, and right inverses. These choices persist for the

smoothly close fields. Transforming each local matrix density back to the product coordinates and summing gives a correction H with $\operatorname{div}(H\nabla u) = r^1$ and $\operatorname{div}(H\nabla v) = r^2$. Thus both fields solve the equations for $A_0 + H$. The fixed bounds, including the finite derivative loss in Lemma 4.3, and the convergence of the sources in every smooth norm give $\|H\|_{L^\infty} \rightarrow 0$ as $T \rightarrow \infty$. These constants are for the fixed end; no estimate uniform in λ or k is used. For large enough T , $A_0 + H$ remains uniformly positive definite.

Return to the original band by setting

$$u_T(T + t, \theta) = T + u(t, \theta), \quad v_T(T + t, \theta) = be^{-\lambda T}v(t, \theta),$$

and translating H to H_T . Linearity preserves both equations. Near the left and right edges, respectively, the pair is the original input and the exact target, and $H_T = 0$. The normal row of A_0 is $(1, 0, 0)$, so the stated positive- t fluxes follow there. Finally, $\partial_t u_T > 0$ by smooth closeness to $U = t$; away from the $2k$ wall graphs, $v_x \neq 0$ in the (s, x, y) coordinates, so the two gradients are independent. This proves the nonzero case.

If v_{in} is identically zero, normalize and interpolate only the first field to t . Its residual is small, compactly supported, and has mean zero because the two mean slopes agree. The compact divergence construction on the connected band gives a small compactly supported flux correction f with $\operatorname{div} f$ equal to this residual. Since $e = \nabla u$ stays nonzero, the symmetric matrix

$$H = \frac{fe^t + ef^t}{|e|^2} - \frac{(f \cdot e)ee^t}{|e|^4}$$

satisfies $He = f$. It is smooth, compactly supported, and small because e stays bounded away from zero on this fixed band. Returning to the original band gives the stated first field, flux, and ellipticity, with $\partial_t u_T > 0$ and the zero field unchanged. $\square$

4.3 Ending a single mode in finite distance

The frequency-changing mechanism below is related to classical nonunique-continuation constructions [9]; we prove all the two-field properties needed here directly. We next keep $u = t$ while terminating the remaining mode. Throughout this construction the normal coefficient is one and the normal-angular cross entries vanish. Only the symmetric angular 2×2 block varies. Thus $u = t$ is always a solution with unit axial flux, and the normal flux of the other field is simply its t derivative.

For a pure mode $f(t) \cos(px)$, a diagonal angular block with x entry

$$a_x(t) = \frac{f''(t)}{p^2 f(t)} > 0 \tag{4.15}$$

solves the equation. The unused diagonal entry can be any fixed positive number. At the end of the matching band, replacing the old constant angular block by this diagonal one preserves the pure mode's equation, because its active entry is unchanged. It also preserves both fields and their normal fluxes at the interface. If necessary, reflect the entire normalized second field on this end, including its retained input and matching region, so that the initial amplitude is positive; include this reflection in the affine normalization that is undone when returning to the original global pair.

Fix a sufficiently large L , to be chosen once below. In a crossing from frequency p in one angular axis to frequency $2p$ in the other, use an interval $|t - t_0| \leq L/p$ and write

$$f(t) = be^{-3p(t-t_0)}, \quad g(t) = be^{-p(t-t_0)}. \tag{4.16}$$

Let $z = p(t - t_0)$. Turn on the second mode by a smooth cutoff that is zero near $z = -L$ and one for $z \geq -L/2$; turn off the first by a cutoff that is one for $z \leq L/2$ and zero near $z = L$. Take these cutoffs with derivatives of order j bounded by fixed constants times L^{-j} . The field is the sum of the two cut-off modes. Before correction the angular diagonal entries are 9 and $1/4$, the squares of the ratios of their decay rates to their frequencies. At least one uncut positive profile remains throughout the crossing: the incoming cutoff is one through $z = L/2$, and the outgoing cutoff is one from $z = -L/2$.

Only one cutoff varies at a time. On such a region, relabel the dominant angular variable as x , its frequency as ℓ , and its amplitude as $D(t) > 0$; label the weaker variable as y , its frequency as m , and its cut-off amplitude as $J(t)$. The residual is $R(t) \cos(my)$. Since the weak-to-dominant ratio is at most e^{-L} on both cutoff regions, and $m/\ell \in \{2, 1/2\}$,

$$\frac{|R|}{\ell^2 D} + \frac{|J|}{D} \leq C e^{-L}. \quad (4.17)$$

Indeed R consists of a second cutoff derivative times the uncut weak profile and twice a first cutoff derivative times its first derivative; scaling by z gives exactly this estimate. The amplitude comparisons are $g/f = e^{2z} \leq e^{-L}$ on the turn-on region and $f/g = e^{-2z} \leq e^{-L}$ on the turn-off region.

The following symmetric angular correction cancels the residual *exactly*:

$$\begin{aligned} H_{xy} = H_{yx} &= \frac{2R}{D\ell m} \sin(\ell x) \sin(my), \\ H_{xx} &= \frac{R}{D\ell^2} \cos(\ell x) \cos(my) - \frac{2RJ}{D^2\ell^2} \sin^2(my), \\ H_{yy} &= 0. \end{aligned} \quad (4.18)$$

To check this, an off-diagonal entry $a \sin(\ell x) \sin(my)$ contributes

$$-aD\ell m \sin^2(\ell x) \cos(my) - aJ\ell m \cos(\ell x) \sin^2(my)$$

to the angular divergence of the flux. The divergence contributions of the two unit xx entries are

$$\begin{aligned} \cos(\ell x) \cos(my) &\mapsto -D\ell^2 \cos(2\ell x) \cos(my), \\ \sin^2(my) &\mapsto -D\ell^2 \cos(\ell x) \sin^2(my). \end{aligned}$$

Substitution of Equation (4.18) leaves just $-R \cos(my)$. Equation (4.17) bounds every correction entry by $C e^{-L}$. Choose L once so large that the perturbed angular block has eigenvalues in a fixed positive interval, independently of p and b , and also that $2e^{-4L} < 1$. No t derivative of this correction occurs in the equation, because its normal row and column are zero.

Between crossings, connect the pure outgoing frequency p at decay rate $p/2$ to the next incoming rate $3p$. Write $r = -f'/f$ and choose a smooth transition, constant near its endpoints, such that

$$p/2 \leq r \leq 3p, \quad 0 \leq r' \leq p^2/8.$$

It fits in an interval of length K/p for one fixed sufficiently large K . Starting from a positive amplitude, the solution $f(t) = f(t_a) \exp(-\int_{t_a}^t r(q) dq)$ stays positive throughout the connector. Equation (4.15) then gives

$$\frac{1}{8} \leq \frac{f''}{p^2 f} = \frac{r^2 - r'}{p^2} \leq 9. \quad (4.19)$$

An initial pure connector joins the fixed starting rate λ at frequency k to $3k$. Choose its positive rate to vary slowly enough that $r^2 - r'$ remains positive. This single connector has finite length and positive finite ellipticity bounds depending on λ/k ; it need not share the later uniform constants. All interfaces preserve the profile and its first derivative, hence the normal flux. Changes in an unused angular entry likewise create no interface source.

Iterate the crossings with frequencies $p_j = 2^j k$, alternating the angular axes. Their lengths and connector lengths form a convergent geometric series. Let b_j be the center amplitude of crossing j , and let $I_j > 0$ be the integral of the decay rate over its following connector. The amplitude at the left edge of crossing j is $b_j e^{3L}$, and its outgoing amplitude at the right edge is $b_j e^{-L}$. Matching to the next crossing therefore gives

$$b_{j+1} = b_j e^{-4L - I_j} \leq b_j e^{-4L}. \quad (4.20)$$

Since $p_j = 2^j k$, this gives the explicit decay

$$p_j b_j \leq k b_0 (2e^{-4L})^j \rightarrow 0.$$

On crossing j and its following connector the profile is bounded by $C b_j e^{3L}$ and its first derivatives by $C p_j b_j e^{3L}$, with fixed constants. The introduced mode, extrapolated backwards from the crossing center, is never larger than the continued dominant mode on its turn-on region. The displayed decay of $p_j b_j$ therefore makes the second field and its entire gradient tend uniformly to zero at the finite limiting section $t = t_*$. Its energy is finite, since the gradient is bounded and the cylinder has finite length.

Extend the second field by zero for $t \geq t_*$. To verify the weak equation, integrate by parts through finitely many stages. Their interface terms cancel by the matching of normal derivatives. At a terminal section tending to t_* the normal flux is $\partial_t v$, which tends uniformly to zero; its contribution against any fixed smooth test function therefore tends to zero. The remaining energy integrals converge by integrability. This proves the weak equation across the accumulation section, with no surface source. The tensor stays bounded and uniformly positive, although its derivatives need not be bounded near t_* . Beyond that section it can be any fixed positive angular block with normal entry one. Add a short final flat band if necessary. On this band the two normalized fields are exactly $u = t$ and $v = 0$.

4.4 Returning to the block and scalarizing

Apply Lemma 4.2 and the ending construction to each of the three ends. If its second field was identically zero, only the one-field matching step and a final flat band are needed. Undo each end's own invertible affine change of fields. The two resulting global fields agree with their original central values and fluxes and have constant terminal voltages and constant angular flux densities. Their integrated terminal currents remain the two original independent slope vectors, multiplied by the common torus area $(2\pi)^2$. In particular those currents are independent and each has zero sum.

Each end now has some finite length $T_i > 0$. For each i separately, map the completed cylinder to its available physical collar by a fixed linear change from $[0, T_i]$ to the collar's transverse interval, followed by the given collar parametrization. Push its tensor density, two fields, and flux pairing forward in the energy form, and retain the central tensor and fields on B_0 . Each collar map is piecewise smooth and bi-Lipschitz with finite nonsingular bounds. These three pushforwards on the collars, together with the central tensor, define a bounded uniformly positive symmetric tensor A_B on the physical block. The bi-Lipschitz constants are fixed after completing the finite cylinders and do not depend on the crossing index.

The fields have finite central and initial collar energy, the retained lengths are finite, and the ending construction has finite energy. Their traces match at each attachment, so matching-trace gluing gives two potentials in $H^1(B)$. For every $\psi \in H^1(B)$, restrict the test to B_0 and the three collars and pull back the collar restrictions. The energy and trace pairings are continuous on the respective H^1 spaces, so the weak identities on the pieces apply to these tests. At each attachment the central and collar outward flux functionals are opposite on the same trace. Summing the identities therefore cancels their terms and leaves only the terminal flux functionals. This is the global weak identity for the glued fields for every such test. Uniform parameter flux density on a terminal torus becomes a constant multiple of its specified measure μ_i : the flux pushforward preserves the flux integral against every pulled-back boundary test function, including through the axial compression. The glued potentials have constant traces on the three boundary components.

We verify all rank and regularity hypotheses of Theorem 3.1. The seams of the collar maps, the finitely many central attachment interfaces, all countably many stage interfaces, including connector interfaces, and the terminal accumulation surfaces have volume zero. Off these sets the tensor and fields are smooth on open pieces. In the central region and unchanged flat ends, the fields are real analytic, being constant-coefficient harmonic functions. Their gradient wedge is either not identically zero on a connected piece, in which case its zero set has measure zero, or is identically zero. In the latter case, on a neighborhood where one gradient is nonzero, write the other field as $v = F(u)$. Its equation gives

$$0 = \operatorname{div}(A\nabla v) = F''(u)\nabla u \cdot A\nabla u,$$

so F is affine there. The critical set of a nonconstant analytic field has measure zero; if both fields are constant there is nothing to check. The elementary analytic zero-set assertion follows by locally differentiating to the first nonzero Taylor term and applying the one-variable zero-set fact along lines and Fubini.

In a matching band with nonzero second input, u is a coordinate and the two gradients are independent off the finitely many smooth walls, by Equation (4.11) and the fixed sign of v_x between them. In a pure or crossing band, $u = t$ and the angular part of the other field is one or two cosines on different axes. Its angular gradient is nonzero off a null set: with one nonzero mode its zeros lie in finitely many walls, and with both modes present they lie in their intersections. Above the accumulation section the second normalized field is zero, an allowed affine dependency. The case with just one matched field has a nonvanishing gradient. These properties are unchanged by invertible affine transformations of the pair or by the physical collar maps. Thus the required regular rank pieces cover B up to a null set.

Theorem 3.1 now provides a bounded positive scalar γ_B and two scalar-conductivity harmonic fields with exactly the same boundary values and normal fluxes as the tensor fields. Together with constants, their boundary values span all separately constant data. Indeed, if a linear combination of the two fields had one common boundary constant, energy minimization for the positive tensor would make it a constant field, so its integrated currents would all be zero. Independence of the two original slope vectors forces that linear combination to be trivial. Their two voltage vectors are therefore independent modulo constants, as required.

Linearity and uniqueness for γ_B now prove Equation (4.1) for every $c \in \mathbb{R}^3$. Testing with $\psi = 1$ gives $\sum_i I_i(c) = 0$. If $I(c) = 0$, test instead with $\psi = U_c$; positivity forces $\nabla U_c = 0$, so connectedness of B makes all c_i equal. Conversely common constant data have zero current. The image therefore has dimension two and is precisely the zero-sum current plane. This proves Proposition 4.1.

5 From a branching block to identical boundary measurements

We now use the exact scalar block supplied by Proposition 4.1. Set $\Omega = B(0, 3)$. Our objective is to construct a uniformly positive bounded scalar conductivity γ , equal to one near the boundary, and a nonzero $w \in H_0^1(\Omega) \cap L^\infty(\Omega)$, supported away from the outer boundary, whose source satisfies

$$\int_{\Omega} \gamma \nabla w \cdot \nabla ((u - u_*)\phi) = 0 \quad (\phi \in C_c^\infty(\Omega))$$

for every bounded γ -harmonic function u , with a constant u_* depending on u . For a sufficiently small nonzero real δ , this identity will make the change from γ to $(1 + \delta w)^2 \gamma$ preserve every boundary measurement. We obtain it by nesting scalar blocks, forcing one common average of u on all cell boundaries, and passing signed surface charges to the limit.

All similarities below act on scalar conductivities by composition, so their ellipticity bounds remain fixed at every depth.

5.1 A repeatable geometry

Set

$$L = \frac{8}{5}, \quad h = \frac{3}{100}, \quad s = \frac{57}{100}, \quad a = \frac{81}{100}. \quad (5.1)$$

The inequalities $2s > 1$ and $2s^3 < 1$ will respectively ensure convergence of the potential series and vanishing volume of the limiting nested set. Define the rectangular annular prism

$$D = \{x \in \mathbb{R}^3 : h < \max(|x_1|/L, |x_2|) < 1, \quad |x_3| < 1\}. \quad (5.2)$$

Let $R(x_1, x_2, x_3) = (x_2, x_1, x_3)$, put

$$S_{\pm}x = sRx \pm ae_1, \quad D_{\pm} = S_{\pm}D, \quad B = D \setminus (\overline{D_-} \cup \overline{D_+}),$$

and write $\Sigma = \partial D$ and $\Sigma_{\pm} = \partial D_{\pm}$.

Lemma 5.1. *The three boundary components of B are disjoint Lipschitz tori. They have compatible piecewise smooth bi-Lipschitz product collars, whose removal leaves a connected Lipschitz central region. Moreover, $\overline{D_-}$ and $\overline{D_+}$ are disjoint compact subsets of D .*

Proof. The outer boxes of the children are, respectively,

$$[-1.38, -0.24] \times [-0.912, 0.912] \times [-0.57, 0.57] \quad \text{and} \quad [0.24, 1.38] \times [-0.912, 0.912] \times [-0.57, 0.57].$$

They lie strictly inside the parent outer box. Both avoid the parent axial hole, whose first-coordinate half-width is $Lh = 0.048$, and the two boxes have separation at least 0.48.

For explicit collars, let $q(\theta) = (q_1(\theta), q_2(\theta))$ parametrize the unit square perimeter by rays, and let $\zeta(\psi)$ similarly parametrize the perimeter of $[h, 1] \times [-1, 1]$ about its center $c_0 = ((1 + h)/2, 0)$. Both parameters have period 2π . Writing

$$(r, z) = c_0 + (1 + \tau)(\zeta(\psi) - c_0),$$

the map

$$(\theta, \psi, \tau) \mapsto (Lrq_1(\theta), rq_2(\theta), z) \quad (5.3)$$

is a product collar of Σ for sufficiently small $|\tau|$. Indeed r stays strictly positive, and on each pair of perimeter faces the map and its inverse have bounded derivatives and nonzero Jacobian. The

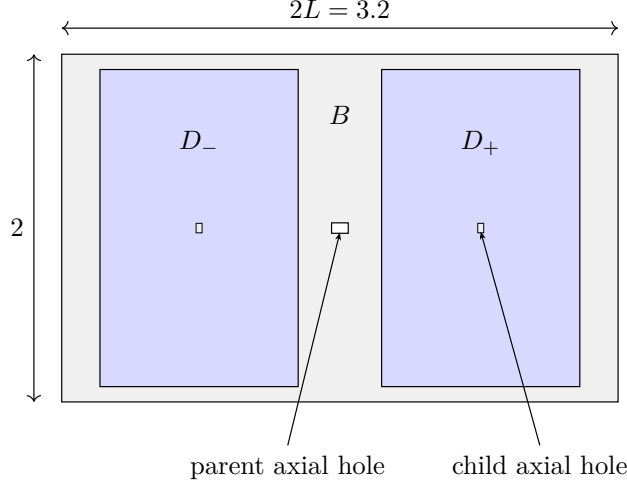


Figure 1: Top view of the cell geometry, at the exact planar scale. The parent prism has $|x_3| < 1$; the two shaded child prisms have $|x_3| < s$. The small gray child holes belong to B , whereas the white central hole is outside D . The available space above and below the children connects the block.

finitely many seams have measure zero. The child collars are its images under S_- and S_+ . Choose their widths small enough that they are disjoint.

For completeness, the central complement is connected. Above the child boxes, for example at height $x_3 = 0.8$, its horizontal section is the connected parent rectangular annulus. A point between heights -0.57 and 0.57 in the complement of the children can move vertically to that level: its horizontal position is outside the child annular footprints or inside one of their axial holes. A point below the child boxes first moves horizontally in the parent annulus to a position outside both boxes and then moves upward. The same argument applies after sufficiently thin collars are removed. Such removal slightly shrinks the parent cross-sectional rectangle in (r, x_3) and slightly expands the two child rectangles; all the containment margins above remain positive.

We also check the local boundary regularity for these particular regions. Positive separation places each boundary point on just one boundary of a rectangular annular prism. A face is locally a half-space, and the convex edge and corner types are graphs of maxima or minima of finitely many affine functions after choosing a diagonal direction. For an inner top corner, the reentrant local model, after changes of coordinates and signs, is $\{\max(x, y) > 0, z < 0\}$. On the line $(x, y, z) = (x_0, y_0, z_0) + t(1, 1, -1)$, membership is equivalent to

$$t > \max\{z_0, -\max(x_0, y_0)\}.$$

The threshold is a Lipschitz function on the transverse plane $x_0 + y_0 - z_0 = 0$. Inner edges give the corresponding two-dimensional max graph, and reversing the side of any of these boundaries reverses the inequality without changing the graph. The remaining edge and corner types give the same max/min description. Thus every boundary point has a local Lipschitz graph. The small collar adjustment keeps this rectangular form and the positive separation, so the central region is Lipschitz as claimed. $\square$

Let μ be the probability measure on Σ obtained by pushing forward $(2\pi)^{-2} d\theta d\psi$ under Equation (5.3) at $\tau = 0$, and let $\mu_{\pm} = (S_{\pm})_*\mu$. These measures have bounded positive densities relative to surface measure on their respective fixed boundaries. Apply Proposition 4.1 to B with

these three measures. We obtain a scalar conductivity b and constants $0 < c_b \leq C_b < \infty$ such that $c_b \leq b \leq C_b$ almost everywhere, with the exact property: every b -harmonic field having separately constant boundary voltages has conormal flux a constant multiple of the corresponding probability measure on each component.

The ball Ω strictly contains $\overline{D}$, since the parent outer box has maximal norm $\sqrt{L^2 + 2} < 3$. For a finite word ν over $\{-, +\}$, let $|\nu|$ denote its length, set $S_\emptyset = I$, and define

$$S_{\nu\pm} = S_\nu \circ S_\pm, \quad D_\nu = S_\nu D, \quad B_\nu = S_\nu B, \quad \Sigma_\nu = \partial D_\nu, \quad \mu_\nu = (S_\nu)_* \mu.$$

Every S_ν is a similarity of ratio $s^{|\nu|}$. Its orthogonal part can reverse orientation, which does not affect any scalar energy or change-of-variables formula. The sets B_ν are pairwise disjoint. Define

$$\gamma(x) = \begin{cases} b(S_\nu^{-1}x), & x \in B_\nu \text{ for some } \nu, \\ 1, & \text{otherwise.} \end{cases} \quad (5.4)$$

This is measurable and satisfies $c \leq \gamma \leq C$, where $c = \min(1, c_b) > 0$ and $C = \max(1, C_b) < \infty$. The remaining nested set has volume zero, since the total volume of the depth- j cells is

$$2^j s^{3j} |D| = (2s^3)^j |D| \longrightarrow 0, \quad 2s^3 = 0.370386 < 1. \quad (5.5)$$

The countably many boundary seams are also null sets. Their assigned values in Equation (5.4) are immaterial.

5.2 Surface charges and equal averages

We first record both the exact source identity and its scaling. For vectors in $\mathbb{R}^3$, $|p|$ denotes Euclidean norm.

Lemma 5.2. *There is a constant K , independent of ν , such that for every $p = (p_0, p_-, p_+)$ with $p_0 + p_- + p_+ = 0$, there exists $W_{\nu,p} \in H_0^1(\Omega) \cap L^\infty(\Omega)$, vanishing off $\overline{D}_\nu$ and constant on each child cell, with*

$$\int_\Omega \gamma \nabla W_{\nu,p} \cdot \nabla \psi = p_0 \int_{\Sigma_\nu} \text{Tr } \psi \, d\mu_\nu + p_- \int_{\Sigma_{\nu-}} \text{Tr } \psi \, d\mu_{\nu-} + p_+ \int_{\Sigma_{\nu+}} \text{Tr } \psi \, d\mu_{\nu+} \quad (5.6)$$

for every $\psi \in H^1(\Omega)$. It obeys

$$\|W_{\nu,p}\|_\infty \leq K s^{-|\nu|} |p|, \quad \int_\Omega |\nabla W_{\nu,p}|^2 \leq K s^{-|\nu|} |p|^2. \quad (5.7)$$

Proof. On the fixed block, let $V = (V_0, V_-, V_+)$ be the constant voltage vector and let p be the integrated outward conormal flux vector of its harmonic extension v . The map $V \mapsto p$ is linear, $\sum_i p_i = 0$, and

$$\int_B b |\nabla v|^2 = \sum_i V_i p_i.$$

Thus its kernel consists exactly of common constants: zero flux implies zero energy, and connectedness of B then makes v constant. The induced map from voltage vectors modulo constants to zero-sum flux vectors is an isomorphism of two-dimensional spaces. Normalize its inverse by $V_0 = 0$.

Extend v by $V_\pm$ in $D_\pm$ and by zero outside $\overline{D}$. Matching traces give an H^1 function W_p with compact support in $\overline{D}$. The exact block flux property gives Equation (5.6) at the root. This identity holds for every H^1 test: equivalently, one may first multiply the test by a smooth cutoff equal to

one near $\overline{D}$. The boundary trace functionals are continuous, since the measures have bounded surface densities. Invertibility of the finite-dimensional map, the maximum principle for the bounded boundary voltages, and the energy identity give

$$\|W_p\|_\infty \leq K |p|, \quad \int |\nabla W_p|^2 \leq K |p|^2.$$

For $j = |\nu|$, set

$$W_{\nu,p}(x) = s^{-j} W_p(S_\nu^{-1}x),$$

where W_p is understood to be zero outside $\overline{D}$. A similarity of ratio r multiplies energy and integrated flux by r in dimension three when the scalar coefficient is copied without changing its values. The extra voltage factor r^{-1} , here $r = s^j$, therefore preserves the prescribed integrated charges and gives the bounds in Equation (5.7). The coefficient inside the children is irrelevant because this function has zero gradient there. Finally, $(S_\nu)_*\mu_\pm = \mu_{\nu\pm}$; hence the measure on each shared boundary is exactly the same from its two adjacent blocks. $\square$

Lemma 5.3. *If $u \in H^1(\Omega)$ is weakly γ -harmonic, then*

$$m_\nu(u) := \int_{\Sigma_\nu} \text{Tr } u \, d\mu_\nu$$

has one common value u_ for all finite words ν . If u is bounded, then $|u_*| \leq \|u\|_\infty$.*

Proof. Test the weak equation for u with $W_{\nu,p}$, and use Equation (5.6) with $\psi = u$. This gives

$$p_0 m_\nu(u) + p_- m_{\nu-}(u) + p_+ m_{\nu+}(u) = 0 \quad \text{whenever } p_0 + p_- + p_+ = 0.$$

The triple of averages is orthogonal to the zero-sum plane and is therefore constant. Iterating proves the assertion. If u is bounded, its traces have the same essential bounds: apply the trace map to the vanishing positive-part truncations beyond those bounds. Since every μ_ν is a probability measure, the last claim follows. $\square$

5.3 A bounded potential carrying opposite limiting charges

Give the two first-level cells charges $q_- = 1$ and $q_+ = -1$, and recursively put $q_{\nu-} = q_{\nu+} = q_\nu/2$. Thus

$$|q_\nu| = 2^{1-j} \quad (|\nu| = j \geq 1), \quad \sum_{|\nu|=j} |q_\nu| = 2. \quad (5.8)$$

Define

$$Z_0 = W_{\emptyset, (0,1,-1)}, \quad Z_j = \sum_{|\nu|=j} W_{\nu, (-q_\nu, q_\nu/2, q_\nu/2)} \quad (j \geq 1), \quad w^{(N)} = \sum_{j=0}^{N-1} Z_j.$$

The source identities telescope. For every integer $N \geq 1$ and every $\psi \in H^1(\Omega)$,

$$\int_\Omega \gamma \nabla w^{(N)} \cdot \nabla \psi = \sum_{|\nu|=N} q_\nu \int_{\Sigma_\nu} \text{Tr } \psi \, d\mu_\nu. \quad (5.9)$$

Lemma 5.4. *The sequence $w^{(N)}$ converges in both $L^\infty(\Omega)$ and $H^1(\Omega)$ to a nonzero $w \in H_0^1(\Omega) \cap L^\infty(\Omega)$ that vanishes almost everywhere off $\overline{D}$.*

Proof. At a fixed depth the cell interiors are disjoint. Combining Equations (5.7) and (5.8), with a larger constant K if necessary, gives

$$\|Z_j\|_\infty \leq K(2s)^{-j}, \quad \|\nabla Z_j\|_2^2 \leq K(2s)^{-j}. \quad (5.10)$$

Since $2s = 1.14 > 1$, both $\sum_j \|Z_j\|_\infty$ and $\sum_j \|\nabla Z_j\|_2$ converge. The first controls the L^2 norms as well because Ω is bounded. This proves the asserted convergence. Every partial sum is in $H_0^1(\Omega)$ and vanishes off $\overline{D}$; the same is true of its limit.

To show that $w \neq 0$, fix $\eta \in C_c^\infty(\Omega)$ equal to one on a neighborhood of $\overline{D_-}$ and zero on a neighborhood of $\overline{D_+}$. The positive separation of these compact sets makes this possible. At every depth the signed total in the first left cell is 1 and that in the first right cell is -1 . Equation (5.9) therefore gives

$$\int_\Omega \gamma \nabla w^{(N)} \cdot \nabla \eta = 1.$$

Strong H^1 convergence yields

$$\int_\Omega \gamma \nabla w \cdot \nabla \eta = 1, \quad (5.11)$$

which proves nontriviality. No trace or pointwise value on the limiting nested set has been used. $\square$

5.4 An exact transformation of all boundary measurements

The transformation in this subsection is related to the conformal energy identity used by Daudé, Kamran, and Nicoleau [5, Proposition 1.2]. Here Equation (5.11) shows that w has a nonzero source. The decisive property is that this source vanishes on every product $(u - u_*)\phi$ with u bounded and harmonic and ϕ a smooth compactly supported test. We prove that identity before transforming the harmonic extensions.

Choose a real $\delta \neq 0$ such that $|\delta| \|w\|_\infty < 1/2$, and set

$$\rho = 1 + \delta w, \quad \tilde{\gamma} = \rho^2 \gamma. \quad (5.12)$$

Then

$$\frac{1}{2} \leq \rho \leq \frac{3}{2}, \quad \frac{c}{4} \leq \tilde{\gamma} \leq \frac{9C}{4} \quad \text{almost everywhere.} \quad (5.13)$$

Both conductivities equal one in a neighborhood of $\partial\Omega$. They differ on a set of positive measure: by Equation (5.11), w is not zero almost everywhere, and positivity of ρ makes $\rho^2 = 1$ equivalent to $w = 0$.

Lemma 5.5. *Let u be a bounded weakly γ -harmonic function in Ω , and let u_* be its common average from Lemma 5.3. Then, for every $\phi \in C_c^\infty(\Omega)$,*

$$\int_\Omega \gamma \nabla \rho \cdot \nabla ((u - u_*)\phi) = 0. \quad (5.14)$$

Proof. The product $(u - u_*)\phi$ belongs to $H_0^1(\Omega)$, so it is an admissible test in Equation (5.9). Put $M = \|u\|_\infty$. For each depth- N cell choose $x_\nu \in \overline{D_\nu}$. The zero-average identity on its surface gives

$$\int_{\Sigma_\nu} (\text{Tr } u - u_*)\phi \, d\mu_\nu = \int_{\Sigma_\nu} (\text{Tr } u - u_*)(\phi - \phi(x_\nu)) \, d\mu_\nu.$$

Since $|\text{Tr } u - u_*| \leq 2M$ almost everywhere on this surface, the absolute value of the sum on the right-hand side of Equation (5.9) is at most

$$4M \|\nabla \phi\|_\infty \text{diam}(D) s^N. \quad (5.15)$$

Here we used Equation (5.8) and $\text{diam}(D_\nu) = s^N \text{diam}(D)$. This tends to zero. The fixed product test has square-integrable gradient, so strong H^1 convergence of $w^{(N)}$, together with boundedness of γ , justifies passage to the limit on the left. Multiplication by δ proves Equation (5.14). $\square$

Proposition 5.6. *The full weak Dirichlet-to-Neumann operators of γ and $\tilde{\gamma}$ on Ω are equal.*

Proof. First take $f \in C^\infty(\partial\Omega)$, and let u be its weak γ -harmonic extension. The weak maximum principle, obtained by testing truncations beyond the boundary bounds, gives $\|u\|_\infty \leq \|f\|_\infty$. Define

$$\tilde{u} = u_* + \frac{u - u_*}{\rho}. \quad (5.16)$$

The Sobolev chain rule applies to $1/\rho$ on the interval $[1/2, 3/2]$, and the product rule applies because both factors are bounded H^1 functions. Hence $\tilde{u} \in H^1(\Omega)$, and it equals u off $\overline{D}$, in particular near the outer boundary. Its flux is

$$\tilde{\gamma} \nabla \tilde{u} = \rho \gamma \nabla u - (u - u_*) \gamma \nabla \rho. \quad (5.17)$$

This is an L^2 vector field.

For $\phi \in C_c^\infty(\Omega)$, use $\rho\phi \in H_0^1(\Omega)$ in the original weak equation to obtain

$$\int_\Omega \rho \gamma \nabla u \cdot \nabla \phi = - \int_\Omega \phi \gamma \nabla u \cdot \nabla \rho.$$

Equation (5.14), expanded by the product rule, also gives

$$\int_\Omega (u - u_*) \gamma \nabla \rho \cdot \nabla \phi = - \int_\Omega \phi \gamma \nabla \rho \cdot \nabla u.$$

Subtracting proves that the divergence of the flux in Equation (5.17) is zero. All cross-gradient products are integrable by Cauchy–Schwarz; boundedness of the other factors also gives the stated L^2 flux bound. Density of smooth tests in $H_0^1(\Omega)$ proves the full weak equation. By uniqueness, $\tilde{u}$ is the $\tilde{\gamma}$ -harmonic extension of f .

Now fix arbitrary $g \in H^{1/2}(\partial\Omega)$ and choose an H^1 extension V of g . Multiply V by a smooth cutoff that equals one near $\partial\Omega$ and is supported in an outer collar disjoint from $\overline{D}$. The resulting V_0 still has trace g , and its gradient is supported where the two solution fluxes agree. Consequently the defining weak pairings satisfy

$$\langle \Lambda_{\tilde{\gamma}} f, g \rangle = \int_\Omega \tilde{\gamma} \nabla \tilde{u} \cdot \nabla V_0 = \int_\Omega \gamma \nabla u \cdot \nabla V_0 = \langle \Lambda_\gamma f, g \rangle.$$

Both operators are bounded from $H^{1/2}$ to $H^{-1/2}$ by the elliptic energy estimate. Since smooth boundary functions are dense in $H^{1/2}(\partial\Omega)$ on the ball, equality extends from smooth f to all boundary traces. This last step does not require extending the transformation in Equation (5.16) to unbounded solutions. $\square$

Proof of Theorem 1.1. Lemma 5.1 permits the use of Proposition 4.1, and Equation (5.4) constructs a uniformly positive bounded scalar conductivity on the ball $\Omega = B(0, 3) \subset \mathbb{R}^3$. Lemmas 5.2–5.4 produce the bounded nonzero function used in Equation (5.12). The second scalar conductivity has the positive bounds in Equation (5.13), agrees with the first near the boundary, and differs from it on a set of positive measure. Set $\gamma_0 = \gamma$ and $\gamma_1 = \tilde{\gamma}$, with common positive finite bounds obtained by taking the smaller lower bound and the larger upper bound. Proposition 5.6 proves equality of their full weak Dirichlet-to-Neumann operators. Thus these coefficients give the claimed counterexample in dimension three, disproving the assertion quantified over all $n \geq 3$. $\square$

Proof of Corollary 1.2. Return to the notation of Corollary 1.2: let Ω be its bounded connected Lipschitz domain, let c, C be the common bounds from Theorem 1.1, and let $B_1, \dots, B_m$ be the prescribed balls. Write $B_i = B(x_i, r_i)$ and set $U_i = B(x_i, r_i/2)$. The closed balls $\overline{U}_i$ are separated from one another and from $\partial\Omega$, even if some prescribed balls have tangent boundaries. Let a_0, a_1 be the pair of Theorem 1.1, and put $\Phi_i(y) = x_i + s_i y$, where $s_i = r_i/6$. This maps $B(0, 3)$ onto U_i . Copy the scalar coefficients by composition: $a_{i,j} = a_j \circ \Phi_i^{-1}$ on U_i . If $q_{i,j}(h)$ is the minimum energy with trace $h \in H^{1/2}(\partial U_i)$, change of variables gives

$$q_{i,j}(h) = s_i \left\langle \Lambda_{a_j}(h \circ \Phi_i), h \circ \Phi_i \right\rangle.$$

Thus $q_{i,0} = q_{i,1}$. The coefficient values and their unit boundary collars are preserved; no amplitude rescaling is needed.

Define $\gamma_\varepsilon = a_{i,\varepsilon_i}$ on U_i and $\gamma_\varepsilon = 1$ elsewhere. The stated bounds follow because the original unit collars imply $c \leq 1 \leq C$. Let $E = \Omega \setminus \bigcup_i \overline{U}_i$, and denote by $E_{\text{ext}}(f, h_1, \dots, h_m)$ the minimum unit-conductivity energy on E with outer trace f and inner traces h_i . The trace theorem and matching-trace H^1 gluing, together with Equation (2.1), give

$$\langle \Lambda_{\gamma_\varepsilon} f, f \rangle = \min_{h_i \in H^{1/2}(\partial U_i)} \left\{ E_{\text{ext}}(f, h_1, \dots, h_m) + \sum_{i=1}^m q_{i,\varepsilon_i}(h_i) \right\} \quad (f \in H^{1/2}(\partial\Omega)).$$

Indeed, restriction of a global competitor gives one inequality, and gluing the subdomain minimizers gives the other; a global minimizer supplies an attaining tuple of interface traces. The right side is independent of ε . Polarization therefore proves equality of the full weak operators.

If two indices differ at position i , their conductivities differ inside U_i on a set of measure $s_i^3 |\{a_0 \neq a_1\}| > 0$. This proves all 2^m choices are distinct. Their common boundary response is not asserted to be that of the constant conductivity one. $\square$

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